

# Optimization Analysis of Whole Genome Sequencing Genotype QC Parameters Tuning Based on Array Genotypes

# Introduction--Background & Purpose (1)

**WGS (Whole Genome Sequencing) vs Array Genotyping:** While WGS can cover the entire genome and discover new variants, the genotype accuracy of individual sites is affected by sequencing depth and algorithms. In contrast, although Array only detects known marker variants, it is commonly used as a "truth" reference for genotypes due to its high precision at individual sites.

**Parameter Optimization Necessity and Key Parameters:** WGS-generated genotypes contain critical quality control parameters. To improve WGS data reliability, we focus on the following 3 parameters:

- **Sequencing Depth (DP):** Represents the total depth at the sequencing site, which is the sum of all allele depths
- **Genotype Quality (GQ):** Calculated based on the PL (Phred-scaled Likelihood) difference between the most likely and second most likely genotypes, used to measure genotype determination confidence
  - GQ=10 corresponds to 90% accuracy
  - GQ=20 corresponds to 99% accuracy
  - GQ=30 corresponds to 99.9% accuracy
- **Allele Fraction (AF):** Represents the proportion of reads supporting the alternate allele at a specific position
  - For heterozygous genotypes, the ideal AF value should be 0.5 (meaning half of the sequencing reads support each allele). However, in practice, due to factors such as allelic preference in sequencing, reference genome bias in alignment algorithms, and mapping quality differences between alleles, the AF value often deviates from the ideal value. By precisely adjusting the AF parameter, false heterozygous genotype calls caused by sequencing errors can be reduced, decreasing false positive genotype calls in association analysis.

## Introduction--Background & Purpose (2)

**Ground Truth Calibration:** Using array data as the reference for true genotypes, we evaluate 4 key **Genotype Consistency Metrics**: **Genotype Concordance**, **Genotype Miss Rate**, **False Positive Rate**, and **False Negative Rate** under different parameter (**DP, GQ, AF**) filter settings by comparing with WGS genotypes, thereby optimizing WGS calling parameters. This parameter optimization analysis based on true genotype data helps establish a more reliable parameter evaluation framework.

DP: Sequencing Depth  
GQ: Genotype Quality  
AF: Allele Fraction

## Genotyping Information

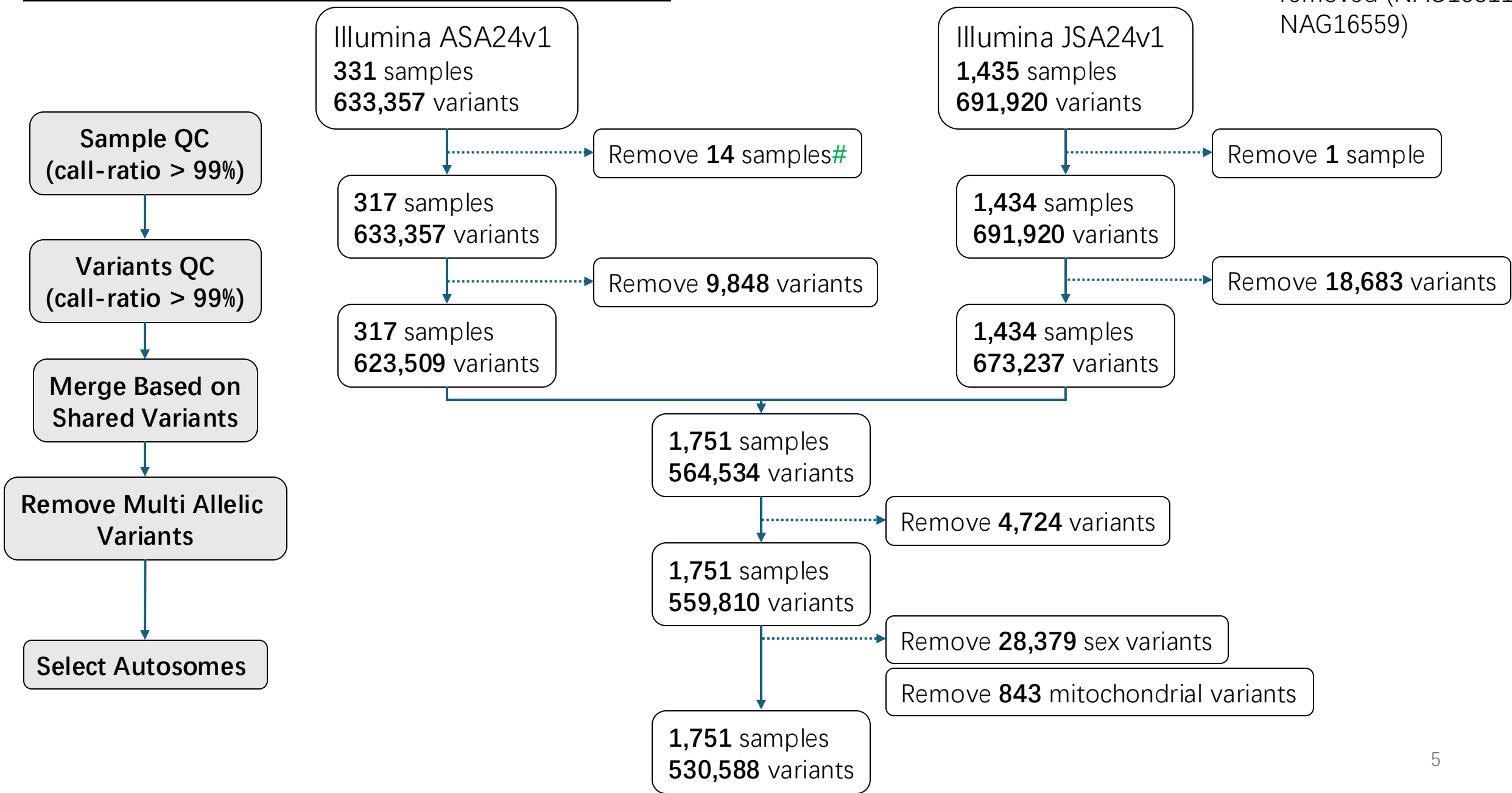
|       |             | Case (CTEPH)                                                             | Control (AGP3K)                                                                                   |
|-------|-------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| WGS   | Sample Size | 390                                                                      | 3135                                                                                              |
|       | Platform    | DNBSeq-G400RS 15x (n=32)<br>NovaSeq 30x (n=26)<br>DNBSeq-T7 30x (n=332)* | HiSeqX 15x                                                                                        |
| Array | Sample Size | 322                                                                      | 1442                                                                                              |
|       | Platform    | Illumina ASA24v1                                                         | Illumina ASA24v1 (n=7)<br>Illumina JSA24v1 (n=1433)<br>Illumina ASA24V1 & Illumina JSA24v1 (n=2)# |

\*: All samples with Array data have corresponding WGS data. Among CTEPH samples, those with both WGS and Array data were all sequenced using DNBSeq-T7 30x.

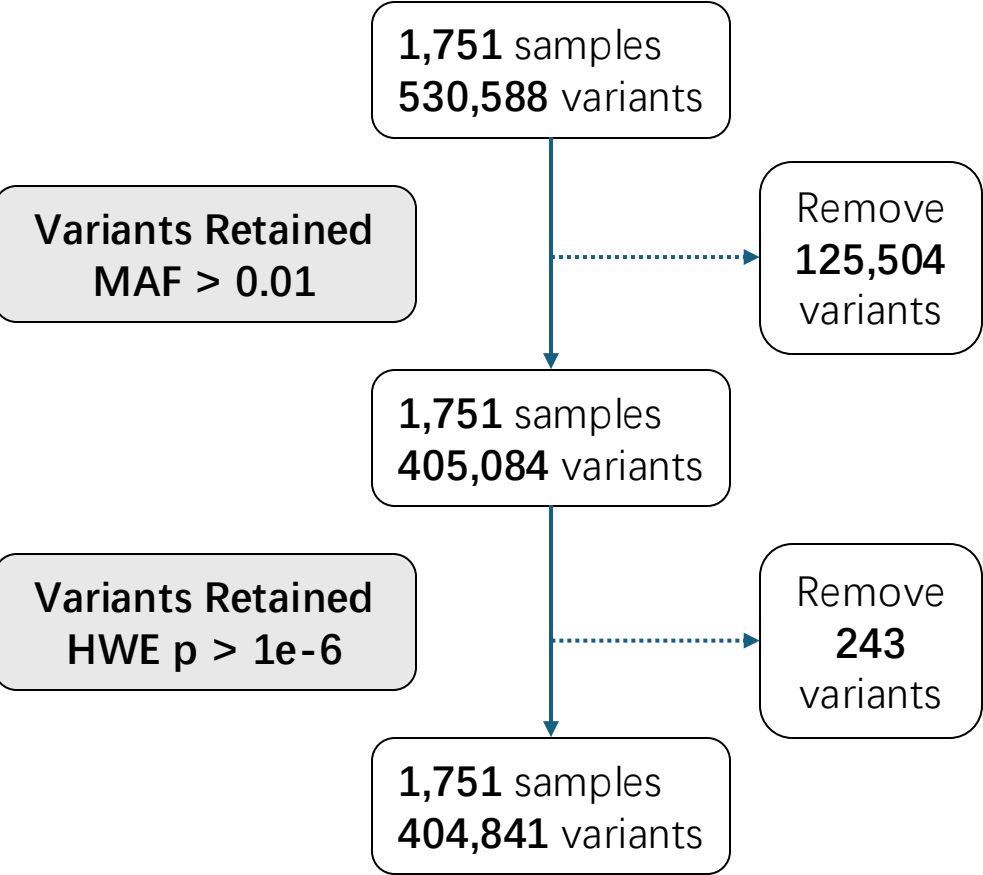
#:2 samples duplicate in Illumina ASA24V1 & Illumina JSA24v1: NAG15811, NAG16559

# Method - True Genotype Prepare (1)

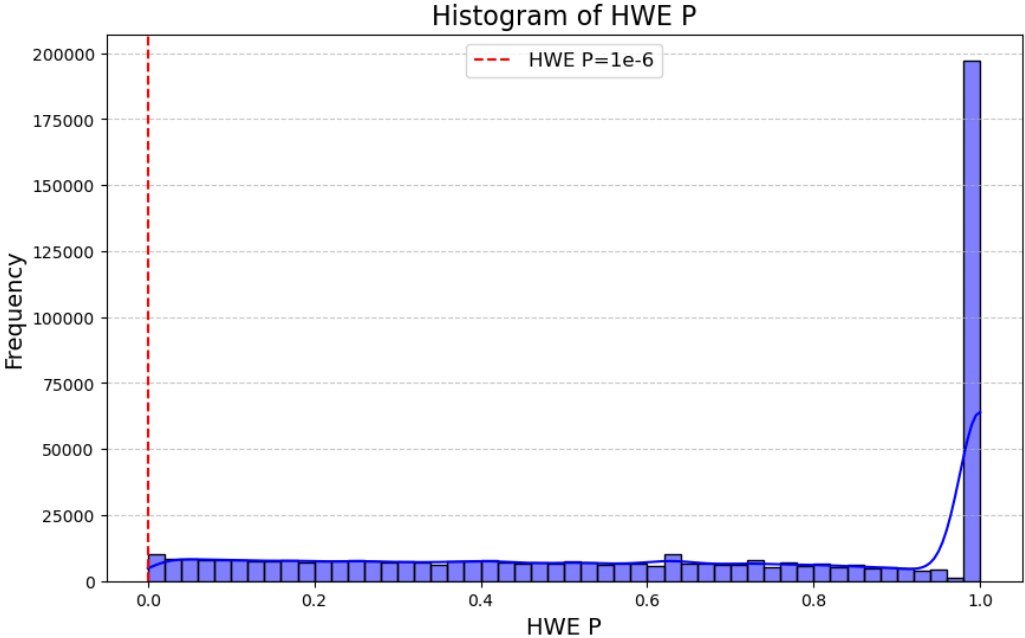
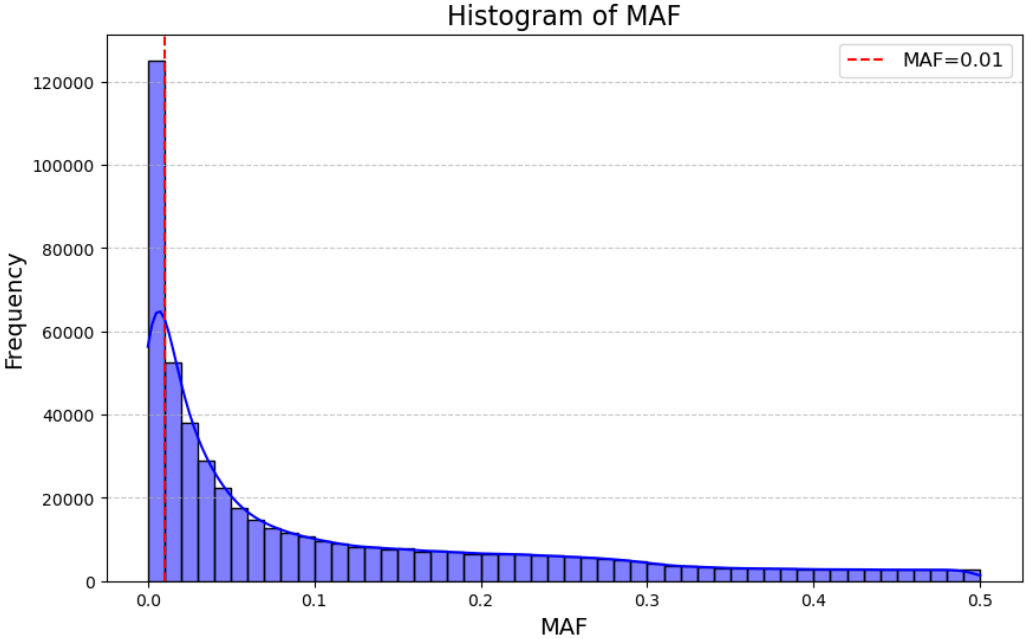
#: 2 duplicate samples removed (NAG15811, NAG16559)



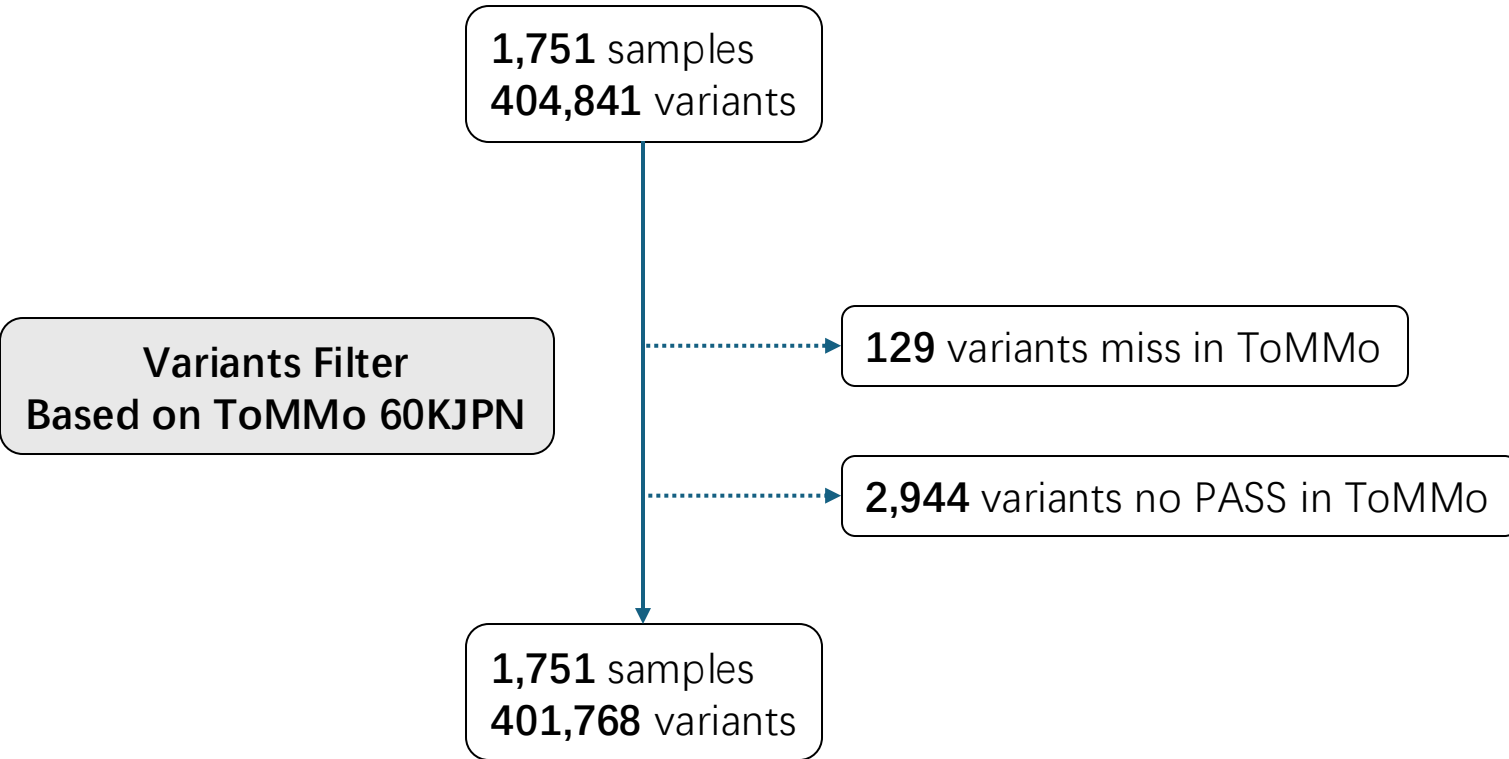
# Method - True Genotype Prepare (2)



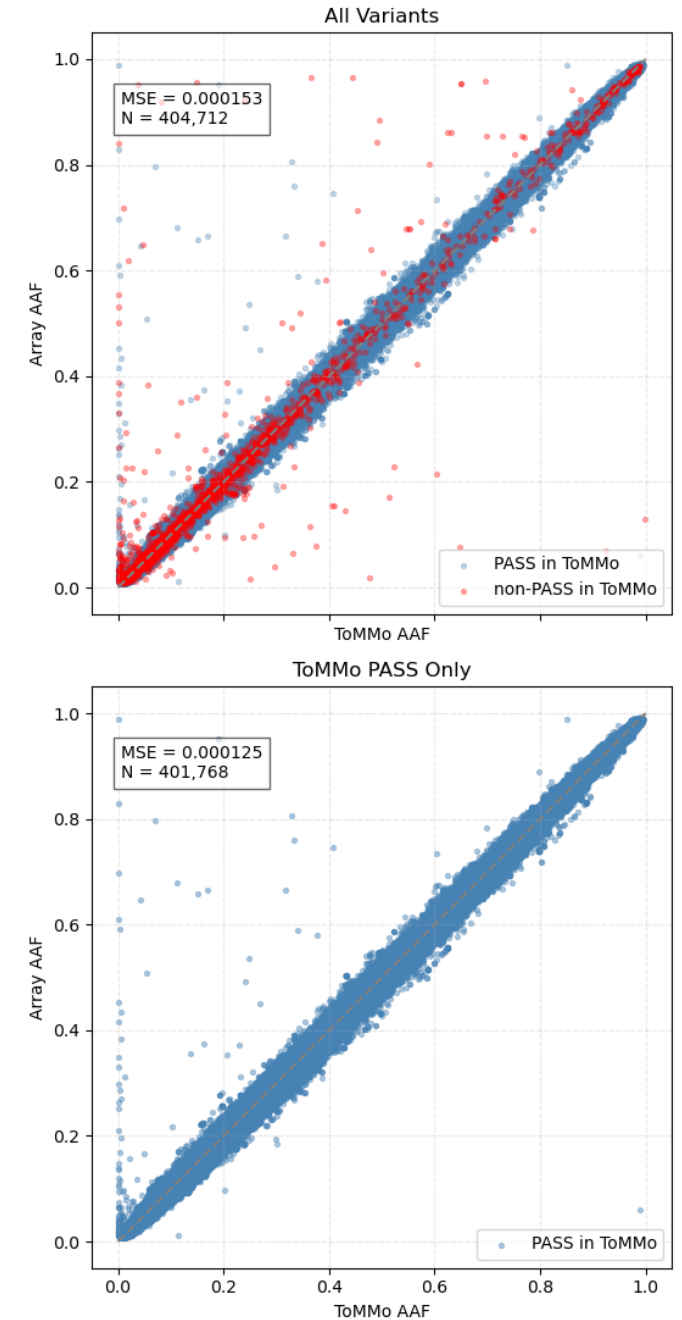
MAF: Minor Allele Frequency  
HWE: Hardy-Weinberg Equilibrium



# Method--True Genotype Prepare (3)



AAF: Alternate Allele Frequency



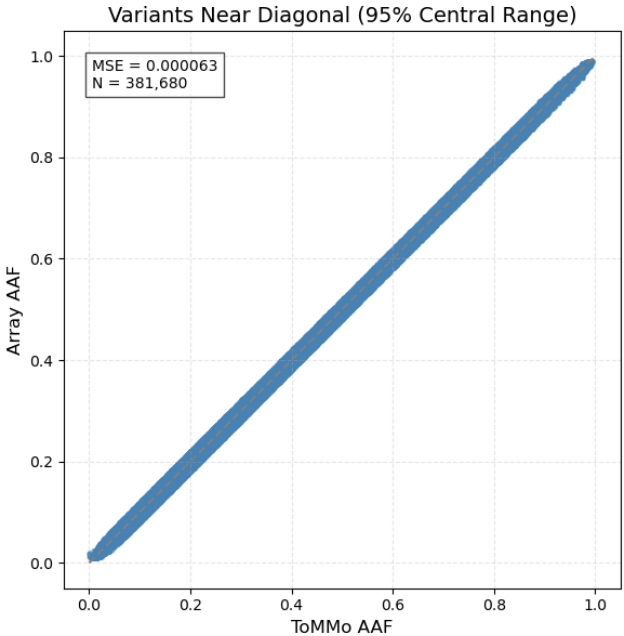
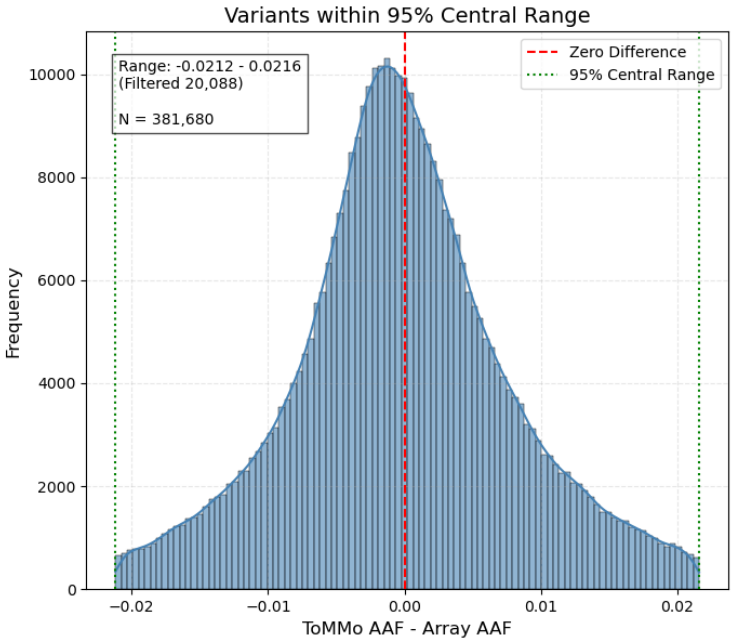
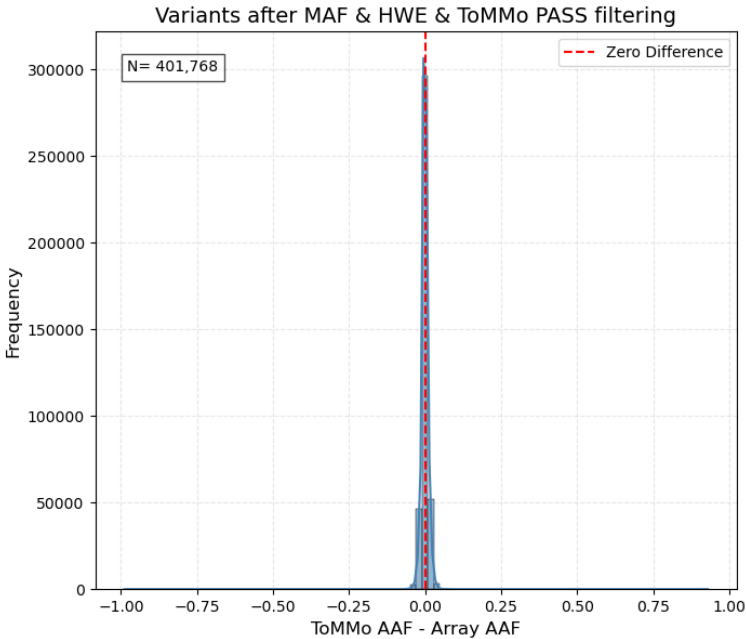
# Method - True Genotype Prepare (4)

1,751 samples  
401,768 variants

Based on ToMMo  
60KJPN, Screen for  
AAF Differential  
Variant Within the  
95% Central Range

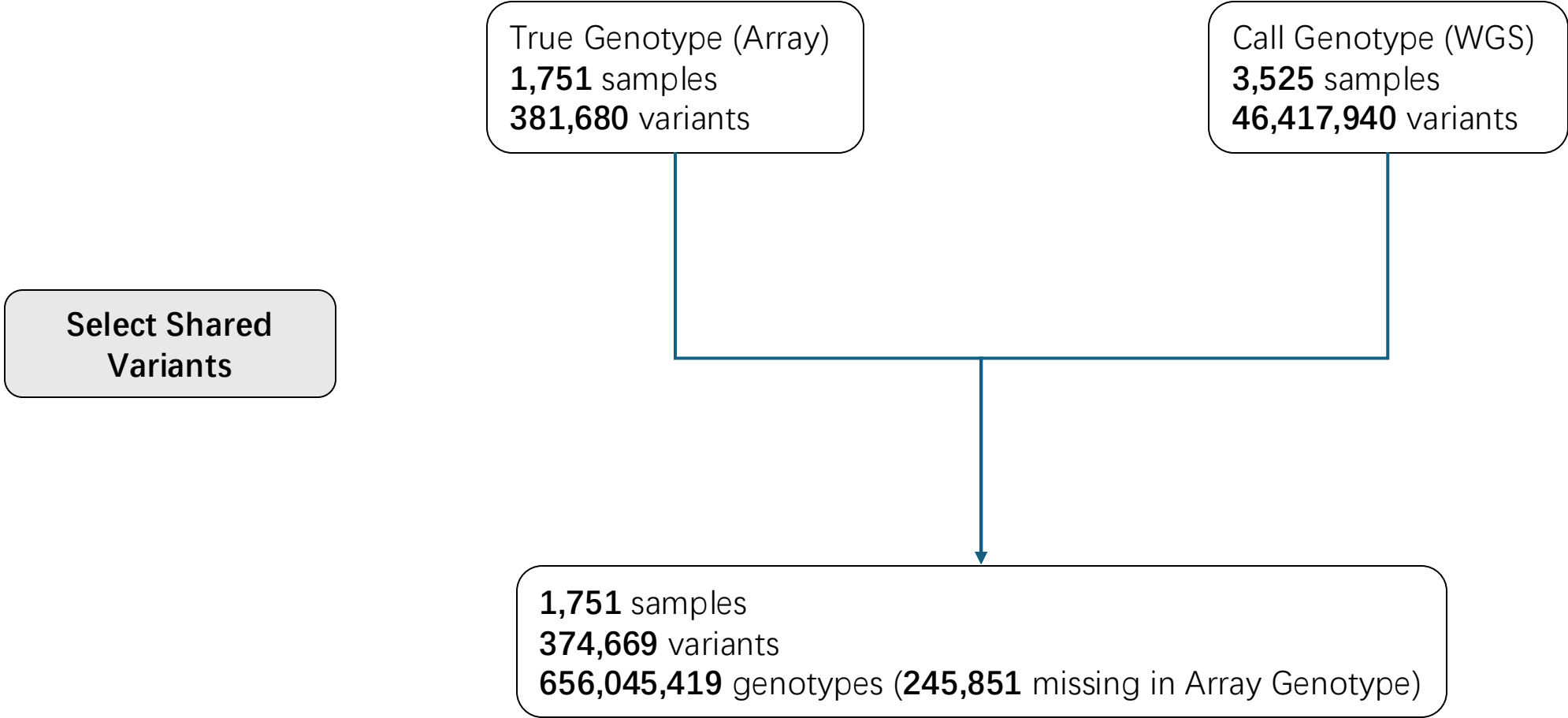
Remove  
20,088  
variants

True Genotype (Array)  
1,751 samples  
381,680 variants

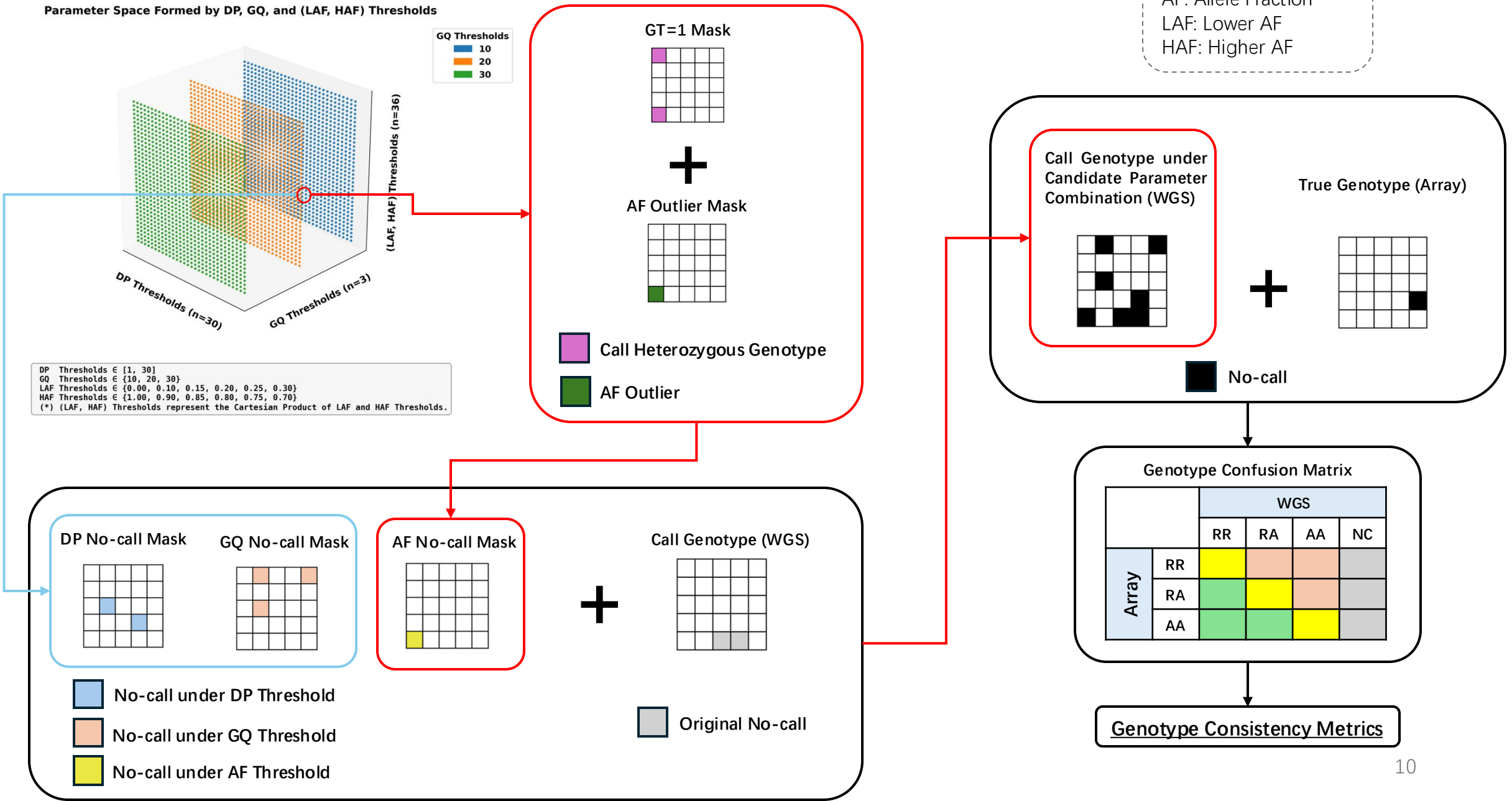


MAF: Minor Allele Frequency  
HWE: Hardy-Weinberg  
Equilibrium  
AAF: Alternate Allele Frequency

# Method--True Genotype & Call Genotype Prepare



# Method - Genotype Consistency Metrics Pipeline



# Method - Genotype Confusion Matrix

|       |           | WGS                |                    |                    |         |
|-------|-----------|--------------------|--------------------|--------------------|---------|
|       |           | Ref / Ref          | Ref / Alt          | Alt / Alt          | No-call |
| Array | Ref / Ref | TP: True Positive  | FP: False Positive | FP: False Positive | Missing |
|       | Ref / Alt | FN: False Negative | TP: True Positive  | FP: False Positive | Missing |
|       | Alt / Alt | FN: False Negative | FN: False Negative | TP: True Positive  | Missing |

FP: False Positive

TP: True Positive

TN: True Negative

FN: False Negative

Missing

Our construction of the **Genotype Confusion Matrix** and **Genotype Consistency Metrics** references the following paper (Kishikawa, Toshihiro, et al. "Empirical evaluation of variant calling accuracy using ultra-deep whole-genome sequencing data." *Scientific reports* 9.1 (2019): 1784.). In WGS data, No-calls include both original missing and genotypes classified as missing under different parameter threshold filtering.

# Method - -Genotype Consistency Metrics (1)

|       |    | WGS |    |    |    |
|-------|----|-----|----|----|----|
|       |    | RR  | RA | AA | NC |
| Array | RR |     |    |    |    |
|       | RA |     |    |    |    |
|       | AA |     |    |    |    |

Genotype  
Concordance

=

|       |    | WGS |    |    |    |
|-------|----|-----|----|----|----|
|       |    | RR  | RA | AA | NC |
| Array | RR |     |    |    |    |
|       | RA |     |    |    |    |
|       | AA |     |    |    |    |

|       |    | WGS |    |    |    |
|-------|----|-----|----|----|----|
|       |    | RR  | RA | AA | NC |
| Array | RR |     |    |    |    |
|       | RA |     |    |    |    |
|       | AA |     |    |    |    |

Genotype  
Miss Rate

=

|       |    | WGS |    |    |    |
|-------|----|-----|----|----|----|
|       |    | RR  | RA | AA | NC |
| Array | RR |     |    |    |    |
|       | RA |     |    |    |    |
|       | AA |     |    |    |    |

# Method - -Genotype Consistency Metrics (2)

|       |    | WGS |    |    |    |
|-------|----|-----|----|----|----|
|       |    | RR  | RA | AA | NC |
| Array | RR |     |    |    |    |
|       | RA |     |    |    |    |
|       | AA |     |    |    |    |

|       |    | WGS |    |    |    |
|-------|----|-----|----|----|----|
|       |    | RR  | RA | AA | NC |
| Array | RR |     |    |    |    |
|       | RA |     |    |    |    |
|       | AA |     |    |    |    |

False Positive Rate =

False Negative Rate =

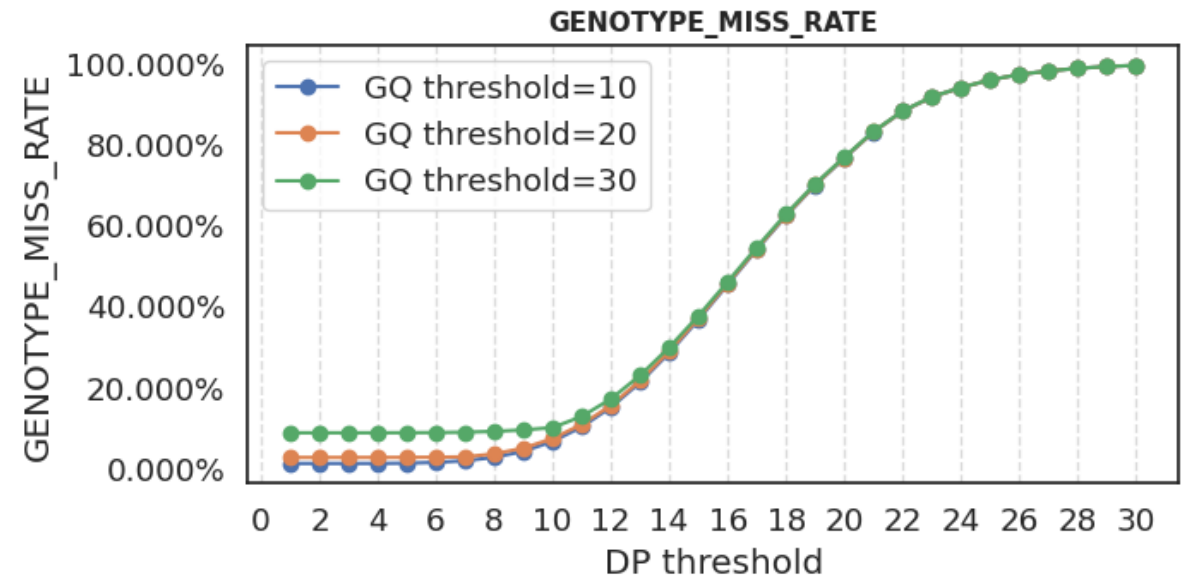
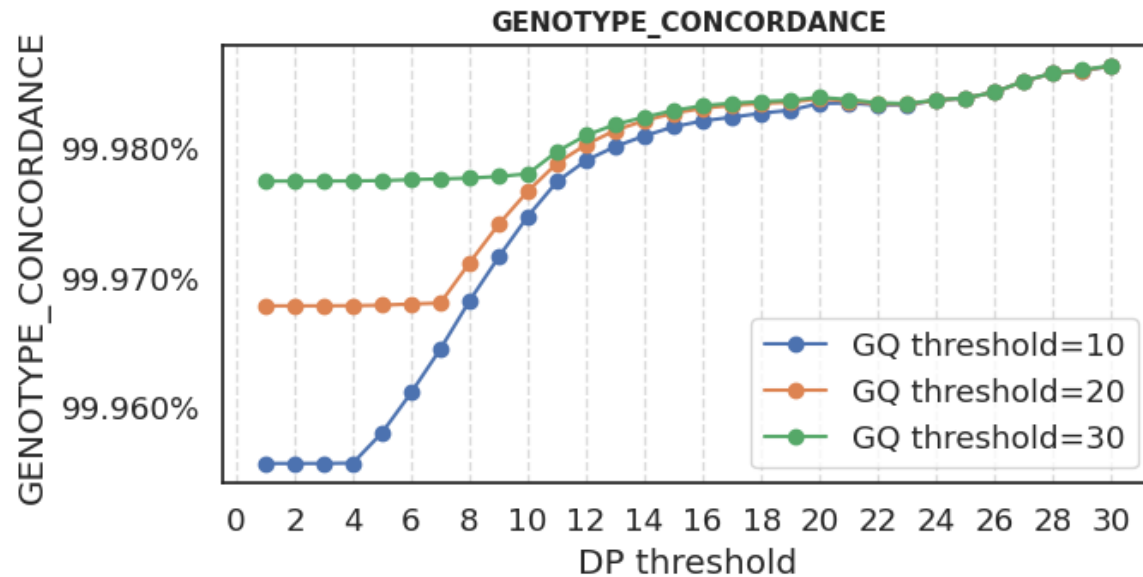
|       |    | WGS |    |    |    |
|-------|----|-----|----|----|----|
|       |    | RR  | RA | AA | NC |
| Array | RR |     |    |    |    |
|       | RA |     |    |    |    |
|       | AA |     |    |    |    |

|       |    | WGS |    |    |    |
|-------|----|-----|----|----|----|
|       |    | RR  | RA | AA | NC |
| Array | RR |     |    |    |    |
|       | RA |     |    |    |    |
|       | AA |     |    |    |    |

# Result--Genotype Consistency Metrics Trends with DP/GQ Threshold Changes (1)

Fix AF Range (0.20, 0.80)

DP & GQ Threshold Effects on Genotype Consistency Metrics in Auto Chr (LAF=0.2, HAF=0.8) — Platform: ALL



## Genotype Concordance

- The concordance rate steadily increases with higher DP thresholds
- Higher GQ thresholds result in better baseline concordance
- All settings converge to nearly 100% concordance at high DP values

## Genotype Miss Rate

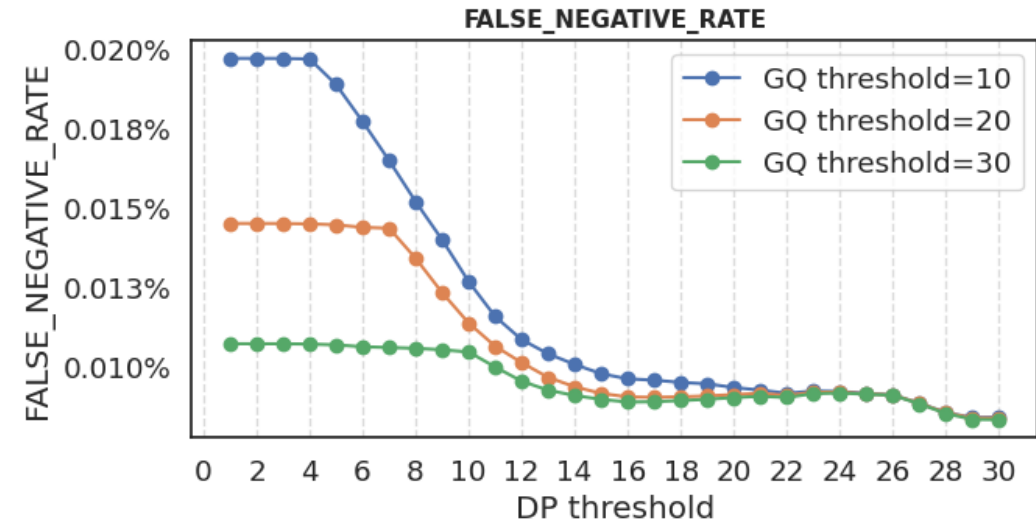
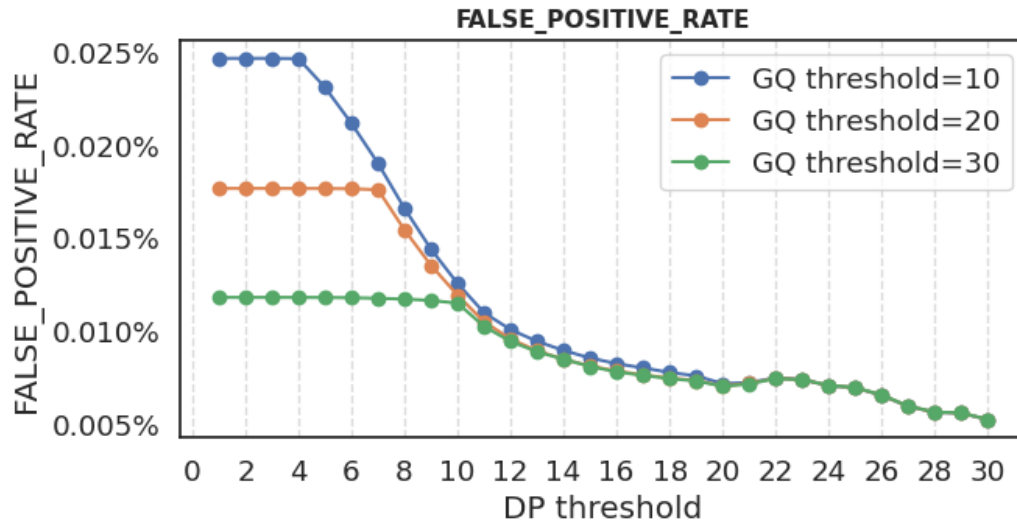
- Remains low at low DP thresholds
- Begins to rise when DP threshold exceeds ~10, eventually reaching about 100%

DP: Sequencing Depth  
GQ: Genotype Quality  
AF: Allele Fraction  
LAF: Lower AF  
HAF: Higher AF

# Result--Genotype Consistency Metrics Trends with DP/GQ Threshold Changes (2)

Fix AF Range (0.20, 0.80)

## DP & GQ Threshold Effects on Genotype Consistency Metrics in Auto Chr (LAF=0.2, HAF=0.8) — Platform: ALL



### False Positive Rate:

- Initial values decrease as GQ threshold increases
- Continuously decreases with increasing DP threshold

### False Negative Rate:

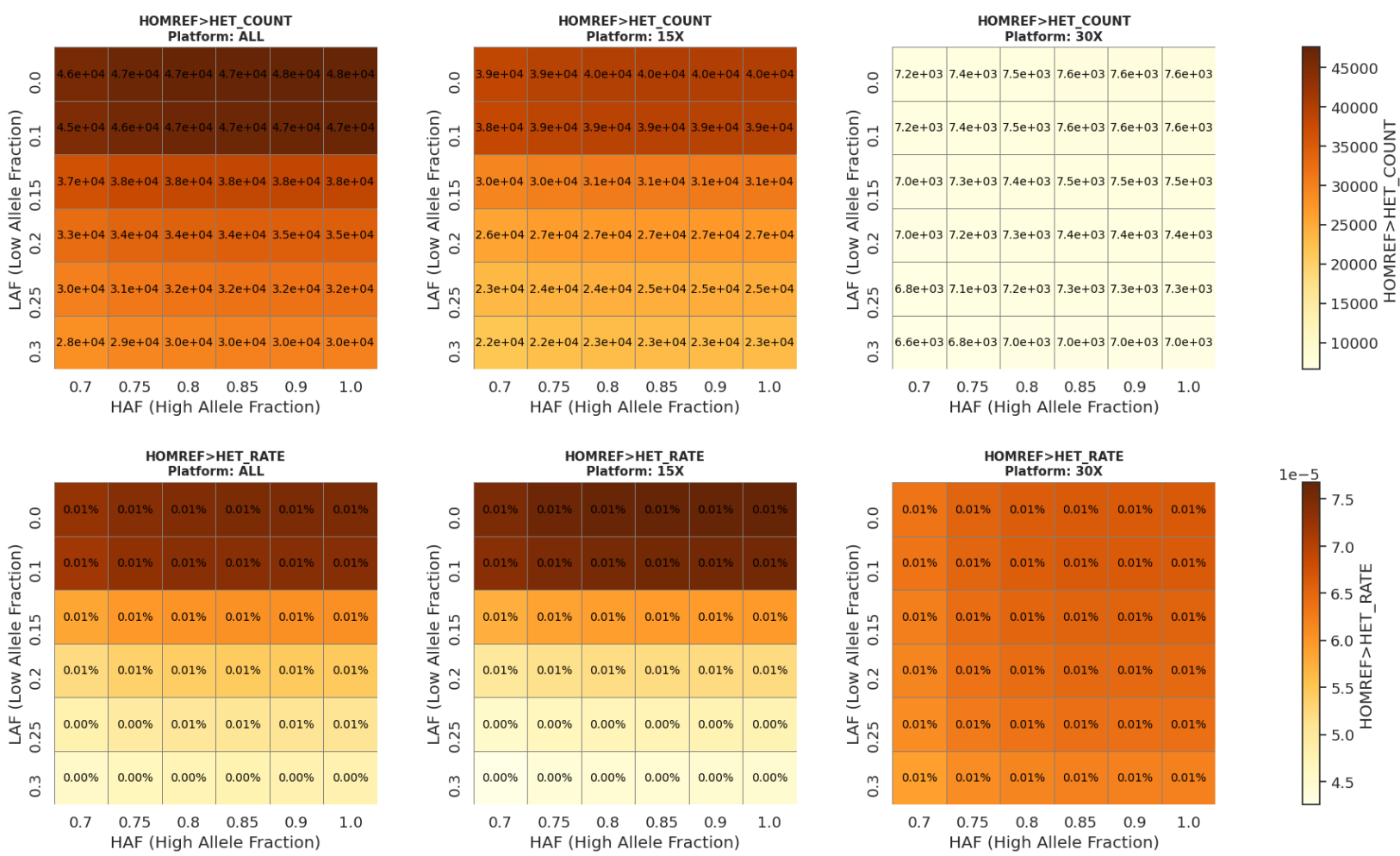
- Shows similar trends to False Positive Rate

DP: Sequencing Depth  
GQ: Genotype Quality  
AF: Allele Fraction  
LAF: Lower AF  
HAF: Higher AF

# Result- -HOMREF>HET Trends with (LAF, HAF) Threshold Changes (1)

Fix DP Threshold = 8; GQ threshold = 20

HOMREF>HET - COUNT and RATE across Platforms (DP=8, GQ=20)



This shows **HOMREF>HET** conversion trends as AF thresholds change:

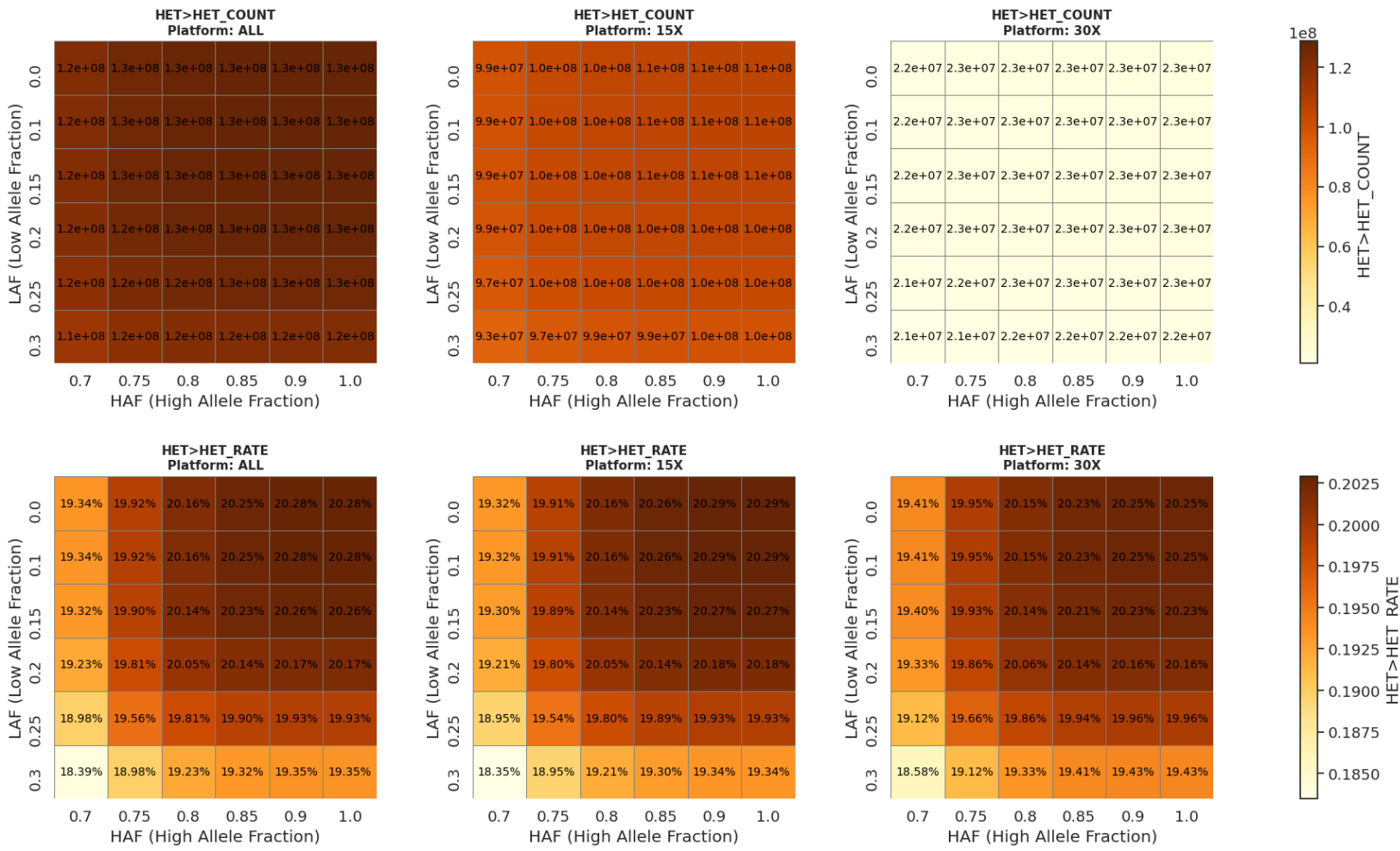
- Stricter AF settings (higher LAF, lower HAF) significantly reduce HOMREF>HET conversion rates
- 30x depth samples show more stable genotyping with threshold changes, while 15x samples have increased misclassification rates with lenient thresholds
- The stricter AF thresholds minimize differences between 15x and 30x depth samples

DP: Sequencing Depth  
GQ: Genotype Quality  
AF: Allele Fraction  
LAF: Lower AF  
HAF: Higher AF

# Result- -HOMREF>HET Trends with (LAF, HAF) Threshold Changes (2)

Fix DP Threshold = 8; GQ threshold = 20

HET>HET - COUNT and RATE across Platforms (DP=8, GQ=20)



Stricter allele fraction (AF) thresholds will reduce the number of **HET>HET** that are correctly identified.

DP: Sequencing Depth  
GQ: Genotype Quality  
AF: Allele Fraction  
LAF: Lower AF  
HAF: Higher AF

# Interpretation--Trade-off Between Accuracy and Data Retention

Overall, these results indicate that higher DP, GQ, AF thresholds provide more accurate genotype calls but also result in more data being filtered out (higher missing rates). In practical applications, a balance must be struck between accuracy and data retention. This leads us to the challenge of **Multi-objective Optimization**.

DP: Sequencing Depth  
GQ: Genotype Quality  
AF: Allele Fraction

# Introduction--Multi-objective Optimization: Overview & Goals

Multi-objective optimization involves finding the best solution when dealing with **multiple competing objectives**. While single-objective problems often have one clear solution, multi-objective problems typically have a set of optimal solutions (Pareto solutions). In these solutions, improving one objective inevitably requires compromising another.

In Genotype Quality Control, we need to balance **2** key metrics (**Genotype Concordance vs. Genotype Miss Rate**), which creates a classic multi-objective optimization challenge:

- Trade-offs exist between objectives—improving concordance leads to higher miss rates
- Metrics are interconnected—changing quality control parameters affects concordance & miss rate simultaneously
- No perfect solution exists—we cannot concordance & miss rate optimize at once

Our goals in this problem are to:

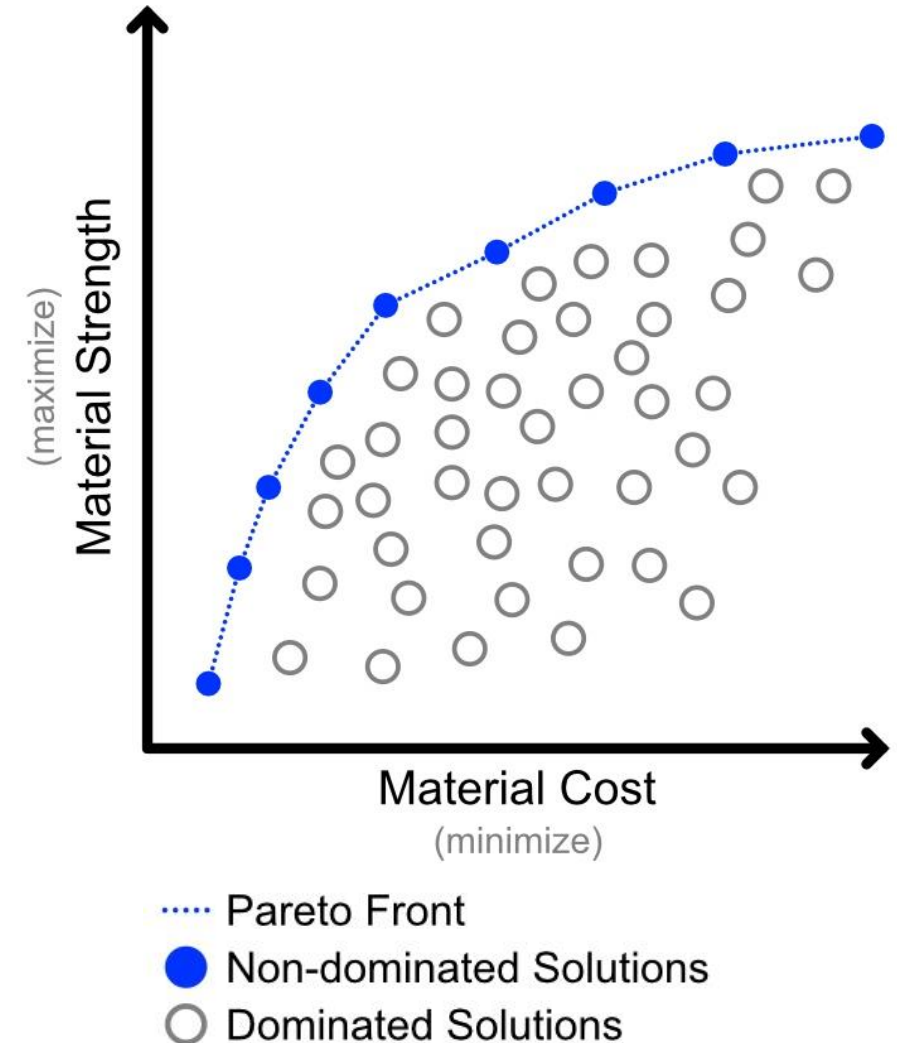
- Find a set of optimal quality control parameters (**Pareto Frontiers**)
- Preserve as much valid data as possible while maintaining data quality
- Provide clear trade-off information for different parameter settings, enabling us to choose parameters that suit our needs

# Method - Pareto Optimization Method (1)

Pareto optimization is a core method for solving multi-objective optimization problems. It is based on these key concepts:

- **Dominated Solution:** Gray points in the figure represent dominated solutions. These points perform worse in both objectives compared to certain points on the Pareto front.
- **Nondominated Solution:** Blue points in the figure represent nondominated solutions. These solutions are not dominated by any other solution, meaning there exists no solution that performs better in all objectives.
- **Pareto Front:** The blue curve represents the Pareto front, showing the set of all **Nondominated Solution**. On this curve, improving one objective necessarily leads to degradation in another objective.

This figure demonstrates a bi-objective optimization problem. Each point on the Pareto front represents an optimal trade-off solution between these two objectives.

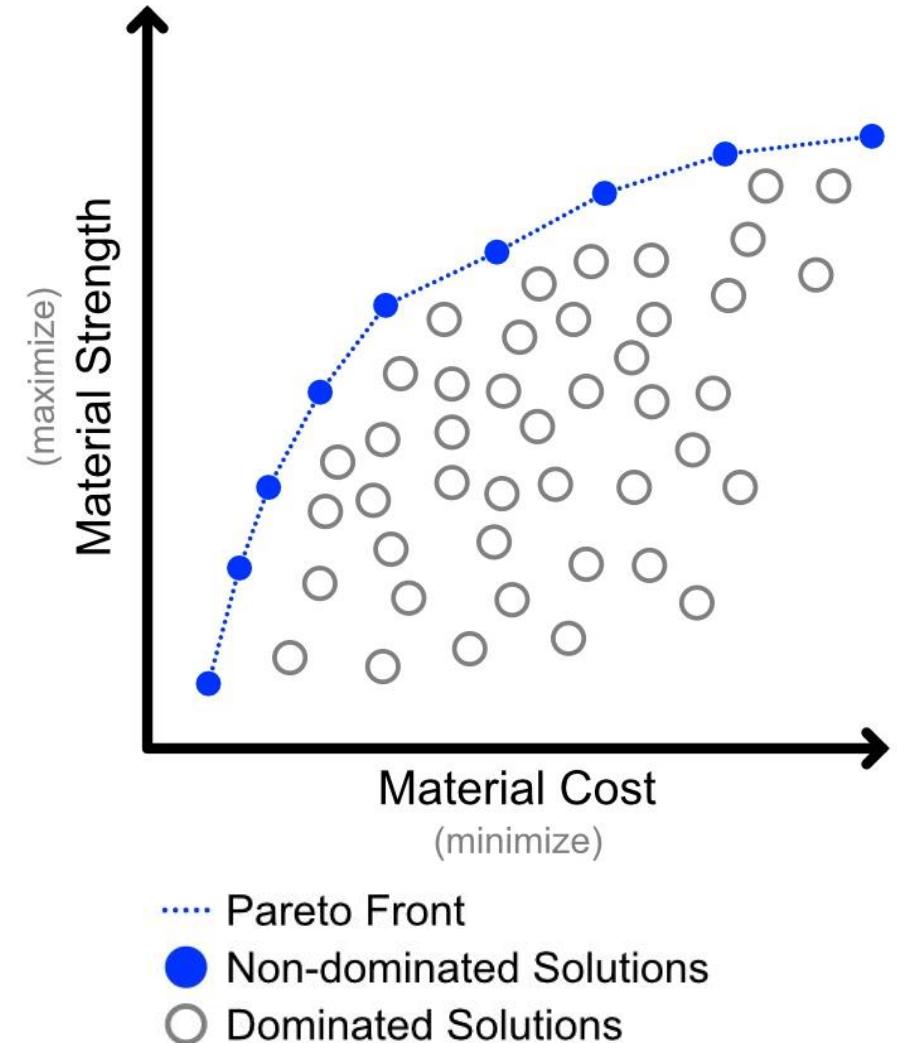


## Method - Pareto Optimization Method (2)

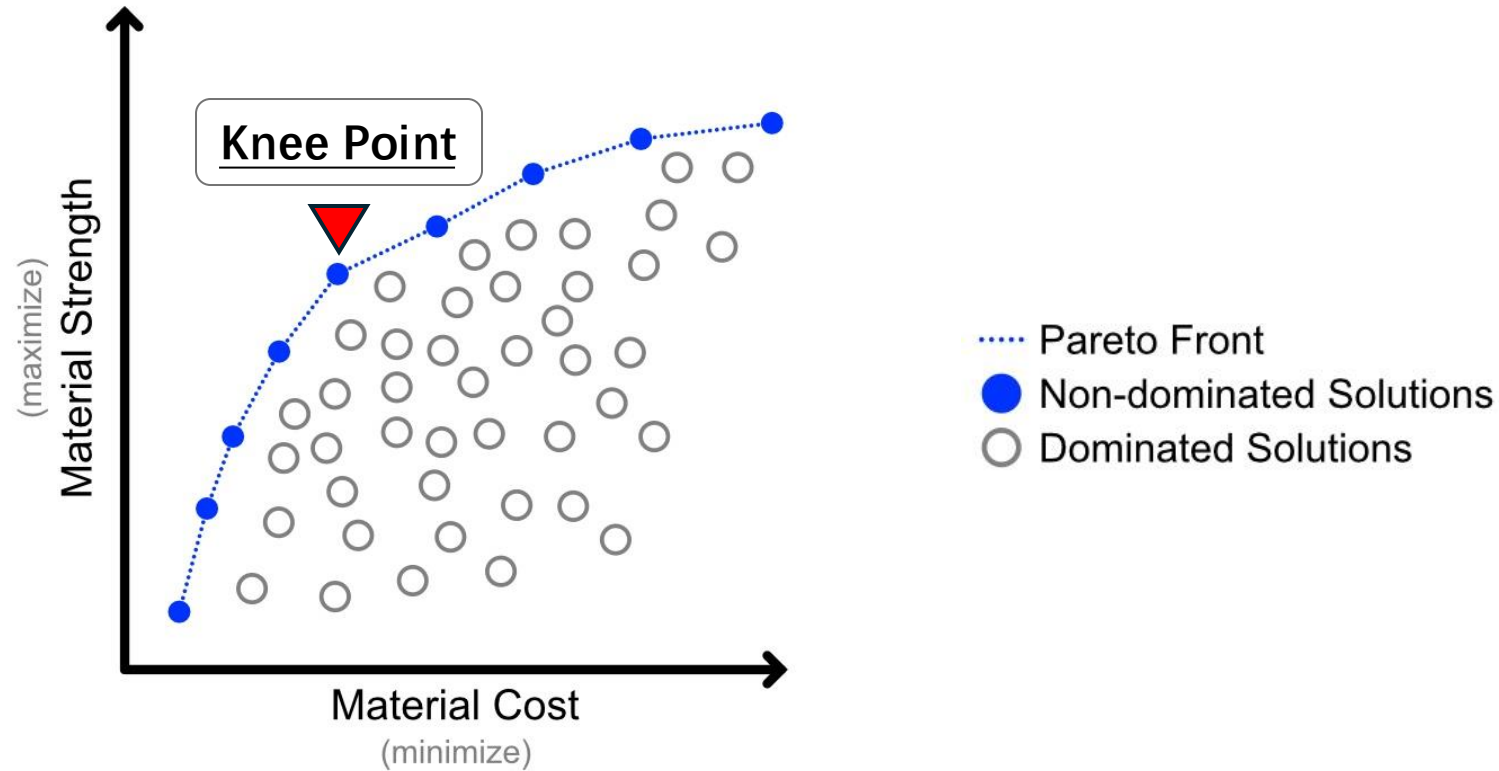
Let's take an example, where x-axis represents **Genotype Miss Rate** and y-axis represents **Genotype Concordance**. Each point represents outcomes from different parameter settings:

- Gray points represent **Dominated Solutions** - meaning there are parameter combinations that can achieve both lower miss rates and higher concordance at the same time
- Blue points are **Nondominated Solutions** on the Pareto Front, which means these points represent the best possible trade-offs between metrics - no point exists that performs better in all metrics (lower miss rate & higher concordance) than this solution (we say no solution dominates it)
- Blue curve is the **Pareto Front**, showing the set of **Nondominated Solutions**. Moving along this line means any improvement in one metric (like increasing concordance) requires sacrificing another metric (increasing miss rate)

Through this approach, we narrow down the selection of optimal parameter combinations to the **Nondominated Solutions on the Pareto Front**.



## Method - Pareto Optimization Method (3)

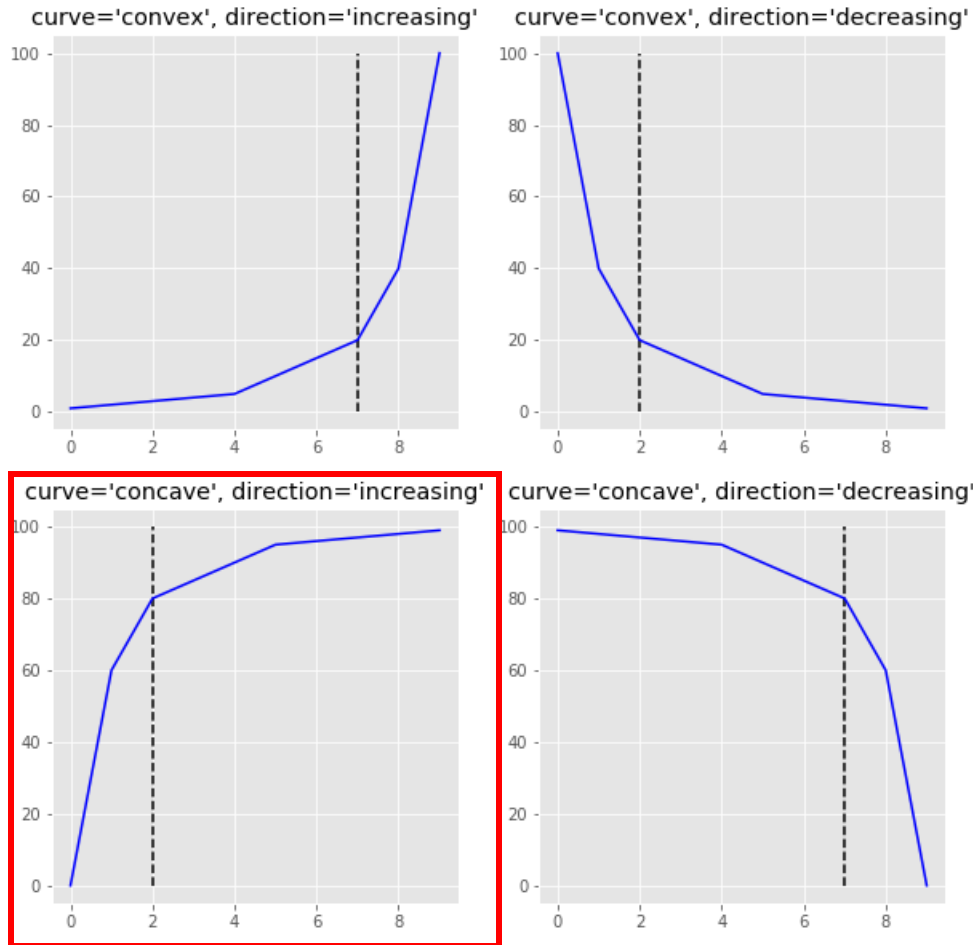


When applying this in practice, we select quality control parameters from the Pareto Front based on research requirements. Specifically:

- On the Pareto Front, we typically use the "**knee point**" as our baseline for these reasons:
  - It offers the most balanced trade-off between all metrics
  - Any attempt to optimize one metric beyond this point severely compromises other metrics
- From this "**knee point**", we can **adjust parameters based on specific needs**. We can choose to emphasize data completeness for large-scale analyses or prioritize genotype accuracy for more precise studies requiring high confidence calls.

This leads us to our next challenges: identifying "**knee point**" and extending the "**weighted methodology**". The weighted methodology allows us to customize parameter optimization by prioritizing specific aspects, while the identification of knee points depends on the scale and dimensions of both x and y variables.

# Introduction--Knee Point Algorithm (Kneed)



## Finding a “Kneedle” in a Haystack: Detecting Knee Points in System Behavior

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<sup>†</sup>Williams College, Williamstown, MA

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<sup>§</sup>International Computer Science Institute, Berkeley, CA

**Abstract**—Computer systems often reach a point at which the relative cost to increase some tunable parameter is no longer worth the corresponding performance benefit. These “knees” typically represent beneficial points that system designers have long selected to best balance inherent trade-offs. While prior work largely uses ad hoc, system-specific approaches to detect knees, we present Kneedle, a general approach to online and offline knee detection that is applicable to a wide range of systems. We define a knee formally for continuous functions using the mathematical concept of curvature and compare our definition against alternatives. We then evaluate Kneedle’s accuracy against existing algorithms on both synthetic and real data sets, and evaluate its performance in two different applications.

encounter knee detection problems similar to those we describe [1], [2], [3], [4], [5]. In these systems, researchers either use ad hoc or system-specific approaches to detect knees, or defer the problem to future work. While a finely-crafted system-specific approach will perform better than a general knee detection approach, a designer may not take the time to design one. Thus, our aim is not to improve or optimize a specific system or protocol, but to provide system designers a general tool for improving the parts of their system they generally do not take the time to optimize. In network protocol and system design, rules-of-thumb often

We use Python kneed library (<https://github.com/arvkevi/kneed?tab=readme-ov-file>) to identify and locate knee points. The core purpose of the kneed algorithm (Kneedle) is to **find the "knee point"** in monotonic curves. Below, I will explain the algorithm implementation using a **Concave Increasing** case as original paper showing.

# Method - Knee Point Algorithm (Kneed) (1)

- **Input Curve**

The input consists of a set of **monotonically increasing** function points. (In our example, x can represent the miss rate, while y can represent concordance rate).

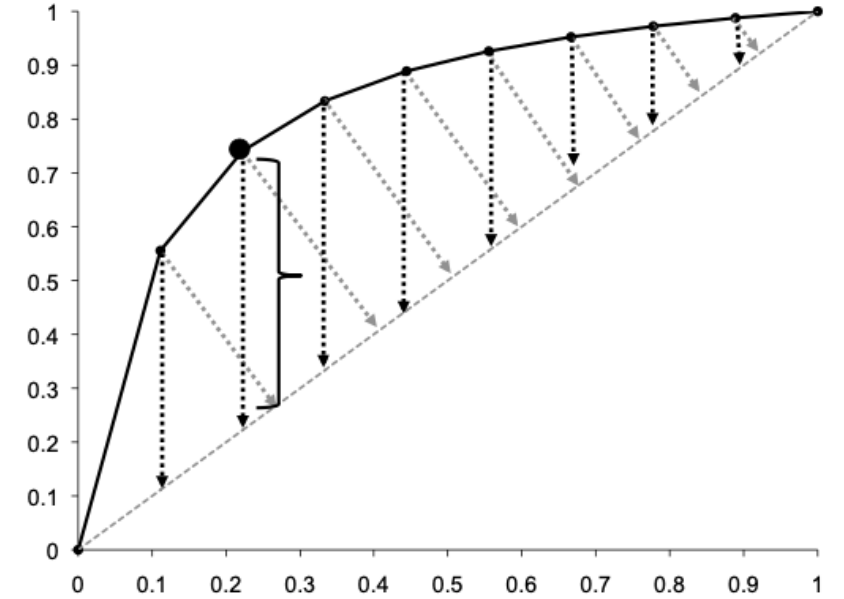
$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$$

- **Smooth Curve Construction**

Through **linear interpolation** ([scipy.interpolate.interp1d](https://docs.scipy.org/doc/scipy/reference/generated/scipy.interpolate.interp1d.html)), kneed obtain a continuous smooth curve  $f(x)$ . This is used to eliminate the interference of data fluctuations in knee point detection.

$$f(x) = \begin{cases} y_i + \frac{y_{i+1} - y_i}{x_{i+1} - x_i} (x - x_i), & x \in [x_i, x_{i+1}] \\ \text{where } i \text{ satisfies } x_i \leq x < x_{i+1} \end{cases}$$

and  $f(x)$  is a monotonically increasing, concave function



- **Normalization**

To make variables of different scales comparable, kneed performs **MinMaxScaler** normalization on the input data  $(x, f(x))$ . This maps the function to the  $[0, 1]$  interval, facilitating subsequent processing.

$$x_{\text{norm}} = \frac{x - \min(x)}{\max(x) - \min(x)}, \quad y_{\text{norm}} = \frac{f(x) - \min(f(x))}{\max(f(x)) - \min(f(x))}$$

- **Difference Curve**

Consider the reference line  $y = x$  (after normalization), which represents linear growth. If a function rises rapidly in its early phase and gradually slows down later, it will be "above" this line in the early phase and "approach" this line in the later phase. Therefore, kneed constructs **Difference Curve** to quantify this process.

$$D(x) = y_{\text{norm}} - x_{\text{norm}}$$

At this point, a larger  $D(x)$  indicates a greater "distance from the reference line ( $y=x$ )," representing the phase of maximum benefit.

# Method - -Knee Point Algorithm (Kneed) (2)

- **Identify Local Extrema as Knee Point Candidates**

On  $D(x)$ , identify all local maximum points. These points indicate: where the current deviation reaches its peak, representing potential **knee point candidates**.

$$D(x_i) \text{ is a local maximum when } D(x_i) \geq D(x_{i-1}) \text{ and } D(x_i) \geq D(x_{i+1})$$

- **Threshold Calculation**

To **prevent oversensitivity** to noise or minor changes (avoiding misidentification of small fluctuations or local variations on the curve as knee points), kneed define a threshold.

$$T(x) = D_{\max} - S \cdot \text{mean}(\Delta x)$$

$D_{\max}$  : local maximum of difference function

$\Delta x$  : step size of normalized  $x$

$S$  : sensitivity parameter determining threshold for "diminishing returns", *default* = 1

- **Knee Point Detection Logic**

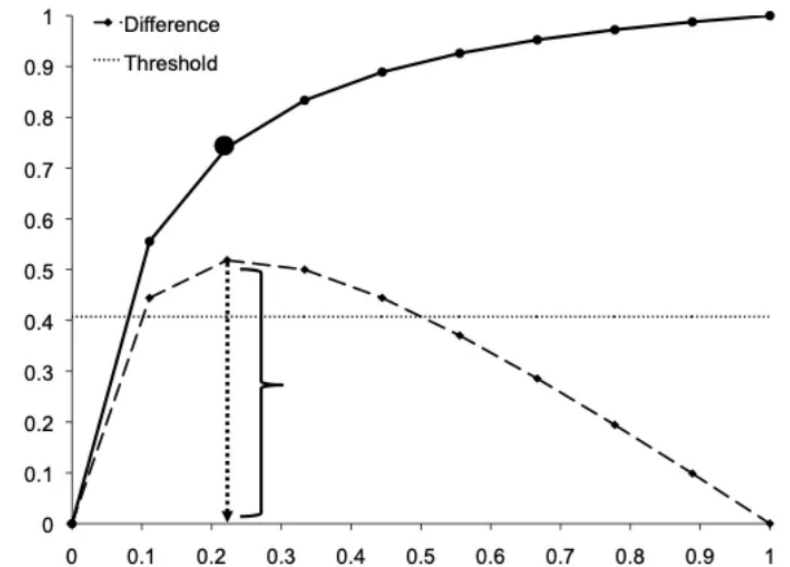
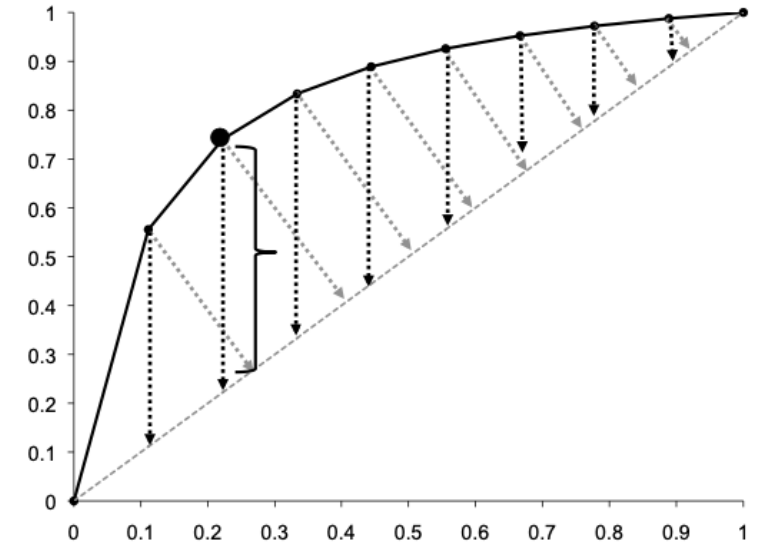
Traversing the difference curve rightward from the first maximum point, when kneed find:

$$D(x_j) > T(x)$$

This indicates:

- The current deviation has increased significantly (exceeding the threshold)
- The function has entered the diminishing returns region

At this point,  $x_j$  is identified as the **knee point**.



# Method - -Extended Kneed Algorithm with Weighted Objectives (1)

The basic Kneed algorithm has limitations when handling multi-objective optimization problems since it assumes all objectives carry equal importance. In real applications, different optimization objectives may have varying weights (such as whether we are willing to accept a higher miss rate to improve concordance rate or reduce false positives and negatives). To address this, we propose a weighted extension version implemented through a **Weighted Difference Curve**.

$$D_w(x) = w_y \cdot y_{\text{norm}}(x) - w_x \cdot x_{\text{norm}}(x)$$

# Method - -Extended Kneed Algorithm with Weighted Objectives (2)

We believe this transformation is valid for the following reasons:

- Mathematical Validity: **Weighted Difference Curve is Generalization of Linear Transformation**

This is a **linear combination transformation** of the original function  $D(x)$ , which can be viewed as a projection from  $x_{norm}, y_{norm}$  space to a scoring space (like PCA projection). Essentially, it modifies how we measure the trade-off between improvements and costs. The linear combination preserves convexity and the existence of extreme points in the space, maintaining the core logic of Kneed's knee point detection.

- Logical Validity: **Simulating Subjective Preferences**

We provide a weight slider to simulate subjective decision preferences (whether one is willing to "pay a higher cost for improvement").

$$D_w(x) = \underbrace{w_y \cdot y_{norm}}_{Benefit} - \underbrace{w_x \cdot x_{norm}}_{Cost}$$

- Geometric Validity: **Transformation of Reference Line Slope**

In the classic definition of Kneed, we compare the curve  $f(x)$  with the line  $y = x$ . Adding weighting factors changes the reference line's slope from 1 to  $\frac{w_y}{w_x}$ .

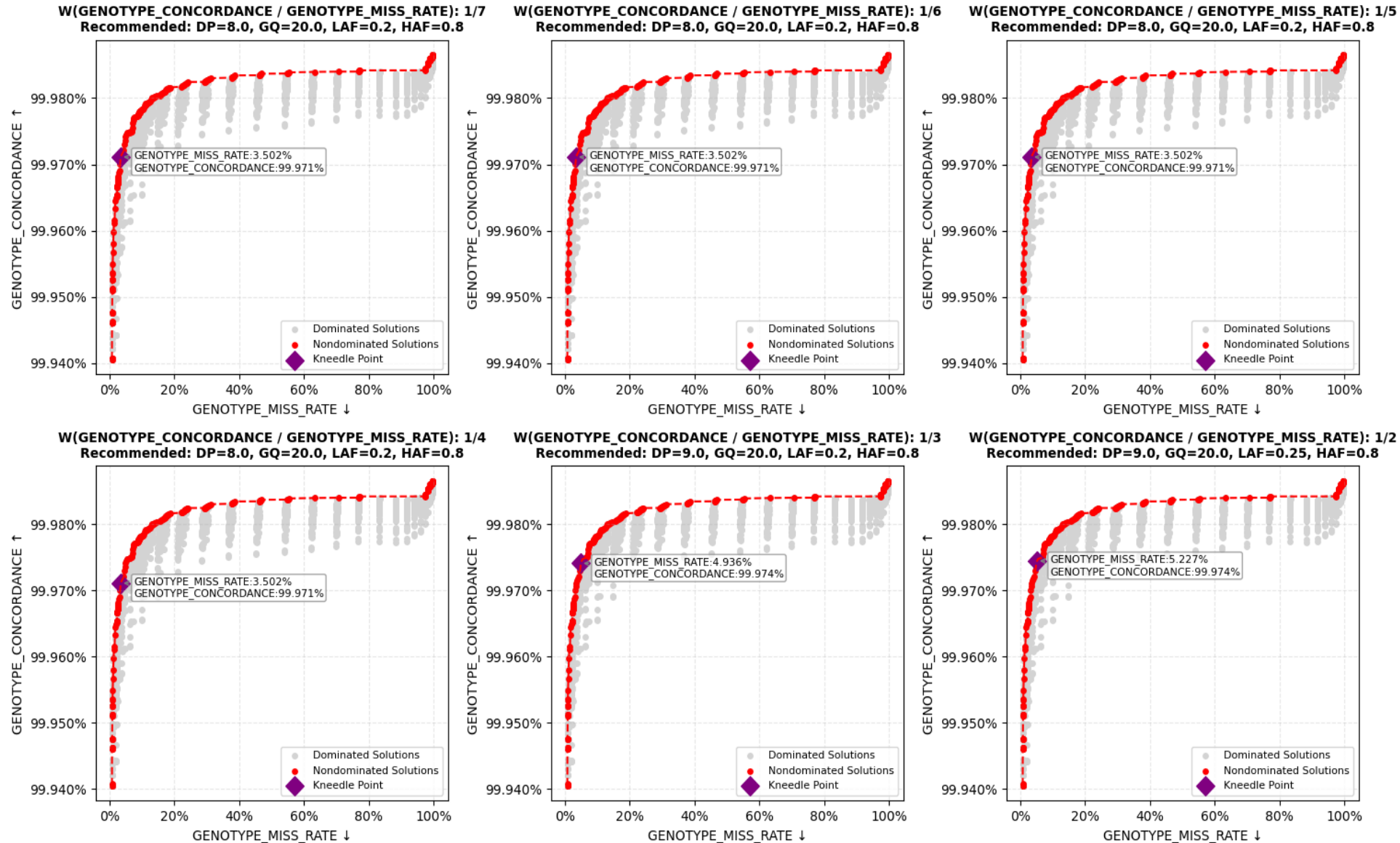
Geometrically, this is equivalent to **tilting the reference direction** around the line with slope 1, making knee point detection more sensitive or conservative.

- When  $w_y > w_x$  the reference line tilts upward, requiring a larger increase in  $x$  to offset an increase in  $y$ . This causes the knee point to **shift rightward**.
- When  $w_x > w_y$ , the reference line tilts downward, causing the knee point to **appear earlier**.

# Result - - Weighted Genotype Consistency Metrics Optimization (1)

DP: Sequencing Depth  
GQ: Genotype Quality  
AF: Allele Fraction  
LAF: Lower AF  
HAF: Higher AF

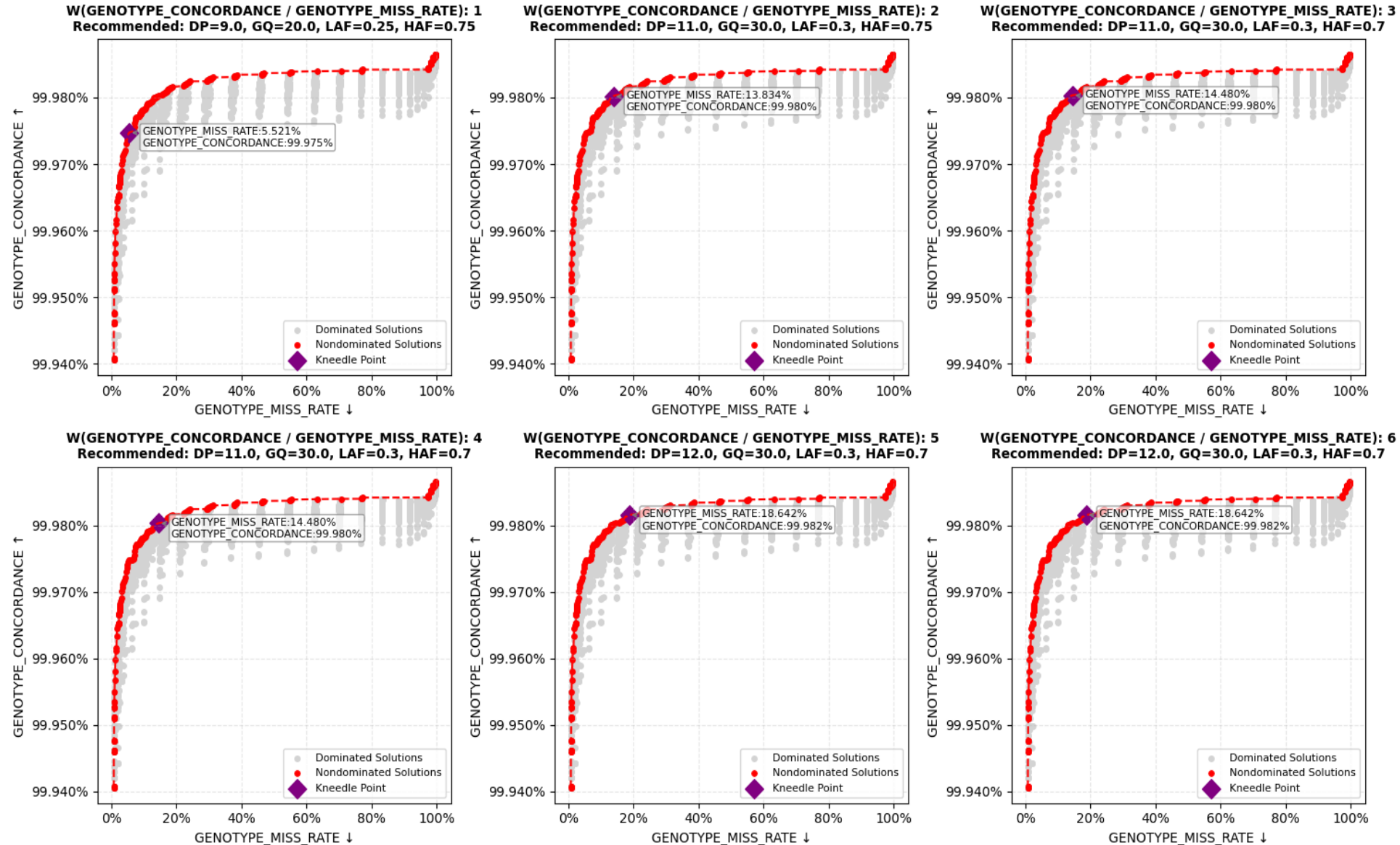
## Trade-off Optimization of Genotype Quality in Auto Chr — Platform:ALL



# Result - - Weighted Genotype Consistency Metrics Optimization (2)

DP: Sequencing Depth  
GQ: Genotype Quality  
AF: Allele Fraction  
LAF: Lower AF  
HAF: Higher AF

## Trade-off Optimization of Genotype Quality in Auto Chr — Platform:ALL



# Result--Weighted Genotype Consistency Metrics Optimization (3)

## **Relationship between weights and recommended parameters:**

- When W (consistency/missing rate) increases from 1:7 to 6:1, the system places more emphasis on consistency
- DP threshold: increases from 8.0 to 12.0, reflecting stricter depth requirements
- GQ threshold: increases from 20.0 to 30.0, indicating higher quality requirements
- AF threshold: LAF increases from 0.2-0.25 to 0.3, HAF decreases from 0.8-0.75 to 0.7, indicating that the Allele Fraction interval for heterozygote determination narrows as weight increases

## **Consistency metric changes:**

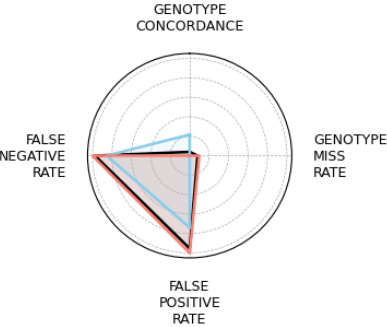
- Genotype consistency: as W value increases, rises from 99.97% to over 99.98%
- Genotype missing rate: as W value increases, rises from 3.5-5% to 18-19%
- Inflection point position moves toward the upper right as W value increases, indicating our algorithm tends to sacrifice data volume for consistency

DP: Sequencing Depth  
GQ: Genotype Quality  
AF: Allele Fraction  
LAF: Lower AF  
HAF: Higher AF

# Result - - Multi-platform Genomic Quality Assessment (1)

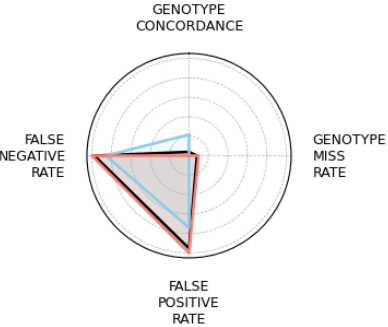
DP: Sequencing Depth  
GQ: Genotype Quality  
AF: Allele Fraction  
LAF: Lower AF  
HAF: Higher AF

W(GENOTYPE\_CONCORDANCE / GENOTYPE\_MISS\_RATE): 1/7  
Recommended: DP=8, GQ=20, LAF=0.20, HAF=0.80



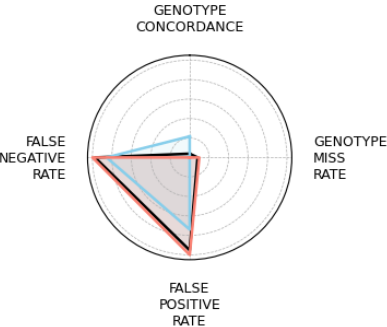
|     | GENOTYPE CONCORDANCE | GENOTYPE MISS RATE | FALSE POSITIVE RATE | FALSE NEGATIVE RATE |
|-----|----------------------|--------------------|---------------------|---------------------|
| ALL | 0.99971              | 0.03502            | 0.00015             | 0.00013             |
| 30X | 0.99973              | 0.02062            | 0.00014             | 0.00013             |
| 15X | 0.99971              | 0.03811            | 0.00016             | 0.00014             |

W(GENOTYPE\_CONCORDANCE / GENOTYPE\_MISS\_RATE): 1/6  
Recommended: DP=8, GQ=20, LAF=0.20, HAF=0.80



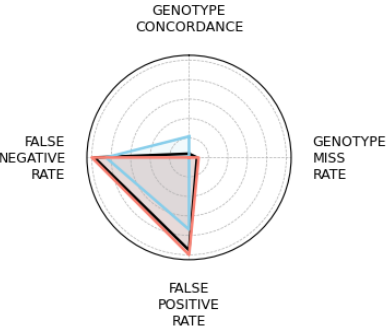
|     | GENOTYPE CONCORDANCE | GENOTYPE MISS RATE | FALSE POSITIVE RATE | FALSE NEGATIVE RATE |
|-----|----------------------|--------------------|---------------------|---------------------|
| ALL | 0.99971              | 0.03502            | 0.00015             | 0.00013             |
| 30X | 0.99973              | 0.02062            | 0.00014             | 0.00013             |
| 15X | 0.99971              | 0.03811            | 0.00016             | 0.00014             |

W(GENOTYPE\_CONCORDANCE / GENOTYPE\_MISS\_RATE): 1/5  
Recommended: DP=8, GQ=20, LAF=0.20, HAF=0.80



|     | GENOTYPE CONCORDANCE | GENOTYPE MISS RATE | FALSE POSITIVE RATE | FALSE NEGATIVE RATE |
|-----|----------------------|--------------------|---------------------|---------------------|
| ALL | 0.99971              | 0.03502            | 0.00015             | 0.00013             |
| 30X | 0.99973              | 0.02062            | 0.00014             | 0.00013             |
| 15X | 0.99971              | 0.03811            | 0.00016             | 0.00014             |

W(GENOTYPE\_CONCORDANCE / GENOTYPE\_MISS\_RATE): 1/4  
Recommended: DP=8, GQ=20, LAF=0.20, HAF=0.80



|     | GENOTYPE CONCORDANCE | GENOTYPE MISS RATE | FALSE POSITIVE RATE | FALSE NEGATIVE RATE |
|-----|----------------------|--------------------|---------------------|---------------------|
| ALL | 0.99971              | 0.03502            | 0.00015             | 0.00013             |
| 30X | 0.99973              | 0.02062            | 0.00014             | 0.00013             |
| 15X | 0.99971              | 0.03811            | 0.00016             | 0.00014             |

## Genomic Quality Evaluation Across ALL / 30X / 15X Platforms Under Varying Trade-off Weights

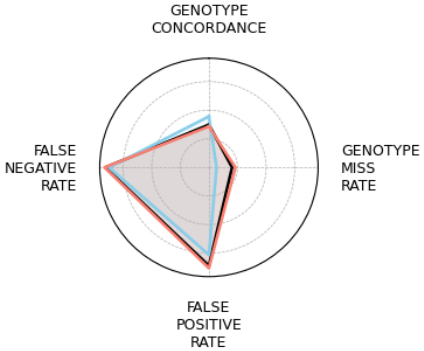
ALL 30X 15X

GENOTYPE CONCORDANCE: 9.9971e-01 ~ 9.9982e-01  
GENOTYPE MISS RATE: 2.0622e-02 ~ 1.9803e-01  
FALSE POSITIVE RATE: 8.6163e-05 ~ 1.5773e-04  
FALSE NEGATIVE RATE: 9.2066e-05 ~ 1.3538e-04

# Result - - Multi-platform Genomic Quality Assessment (2)

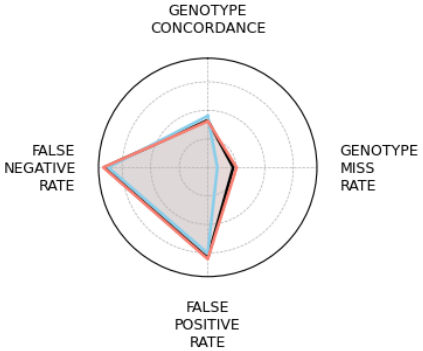
DP: Sequencing Depth  
GQ: Genotype Quality  
AF: Allele Fraction  
LAF: Lower AF  
HAF: Higher AF

W(GENOTYPE\_CONCORDANCE / GENOTYPE\_MISS\_RATE): 1/3  
Recommended: DP=9, GQ=20, LAF=0.20, HAF=0.80



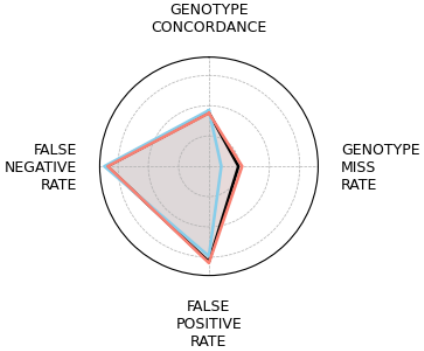
|     | GENOTYPE CONCORDANCE | GENOTYPE MISS RATE | FALSE POSITIVE RATE | FALSE NEGATIVE RATE |
|-----|----------------------|--------------------|---------------------|---------------------|
| ALL | 0.99974              | 0.04936            | 0.00014             | 0.00012             |
| 30X | 0.99975              | 0.03014            | 0.00013             | 0.00012             |
| 15X | 0.99974              | 0.05349            | 0.00014             | 0.00012             |

W(GENOTYPE\_CONCORDANCE / GENOTYPE\_MISS\_RATE): 1/2  
Recommended: DP=9, GQ=20, LAF=0.25, HAF=0.80



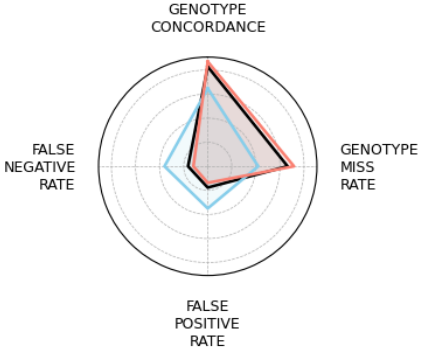
|     | GENOTYPE CONCORDANCE | GENOTYPE MISS RATE | FALSE POSITIVE RATE | FALSE NEGATIVE RATE |
|-----|----------------------|--------------------|---------------------|---------------------|
| ALL | 0.99974              | 0.05227            | 0.00013             | 0.00012             |
| 30X | 0.99975              | 0.03261            | 0.00013             | 0.00012             |
| 15X | 0.99974              | 0.05650            | 0.00013             | 0.00012             |

W(GENOTYPE\_CONCORDANCE / GENOTYPE\_MISS\_RATE): 1  
Recommended: DP=9, GQ=20, LAF=0.25, HAF=0.75



|     | GENOTYPE CONCORDANCE | GENOTYPE MISS RATE | FALSE POSITIVE RATE | FALSE NEGATIVE RATE |
|-----|----------------------|--------------------|---------------------|---------------------|
| ALL | 0.99975              | 0.05521            | 0.00013             | 0.00012             |
| 30X | 0.99975              | 0.03513            | 0.00013             | 0.00012             |
| 15X | 0.99975              | 0.05953            | 0.00013             | 0.00012             |

W(GENOTYPE\_CONCORDANCE / GENOTYPE\_MISS\_RATE): 2  
Recommended: DP=11, GQ=30, LAF=0.30, HAF=0.75



|     | GENOTYPE CONCORDANCE | GENOTYPE MISS RATE | FALSE POSITIVE RATE | FALSE NEGATIVE RATE |
|-----|----------------------|--------------------|---------------------|---------------------|
| ALL | 0.99980              | 0.13834            | 0.00010             | 0.00010             |
| 30X | 0.99978              | 0.09502            | 0.00011             | 0.00011             |
| 15X | 0.99981              | 0.14766            | 0.00010             | 0.00010             |

## Genomic Quality Evaluation Across ALL / 30X / 15X Platforms Under Varying Trade-off Weights

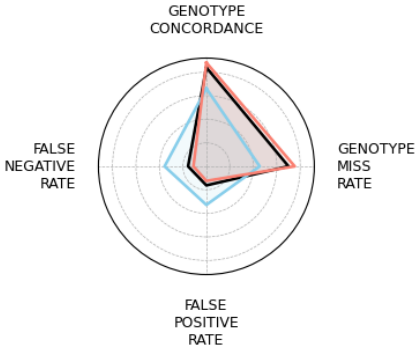
ALL 30X 15X

GENOTYPE CONCORDANCE: 9.9971e-01 ~ 9.9982e-01  
GENOTYPE MISS RATE: 2.0622e-02 ~ 1.9803e-01  
FALSE POSITIVE RATE: 8.6163e-05 ~ 1.5773e-04  
FALSE NEGATIVE RATE: 9.2066e-05 ~ 1.3538e-04

# Result - - Multi-platform Genomic Quality Assessment (3)

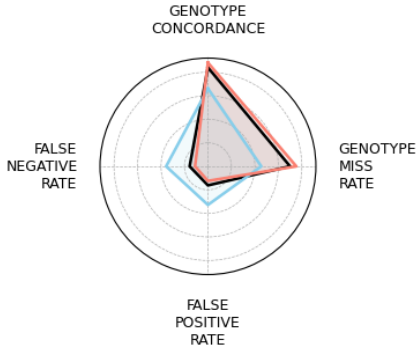
DP: Sequencing Depth  
GQ: Genotype Quality  
AF: Allele Fraction  
LAF: Lower AF  
HAF: Higher AF

W(GENOTYPE\_CONCORDANCE / GENOTYPE\_MISS\_RATE): 3  
Recommended: DP=11, GQ=30, LAF=0.30, HAF=0.70



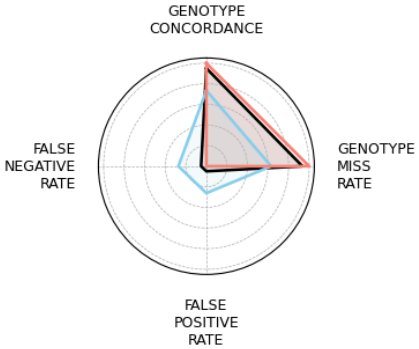
|     | GENOTYPE CONCORDANCE | GENOTYPE MISS RATE | FALSE POSITIVE RATE | FALSE NEGATIVE RATE |
|-----|----------------------|--------------------|---------------------|---------------------|
| ALL | 0.99980              | 0.14480            | 0.00010             | 0.00010             |
| 30X | 0.99978              | 0.10124            | 0.00011             | 0.00011             |
| 15X | 0.99981              | 0.15418            | 0.00010             | 0.00010             |

W(GENOTYPE\_CONCORDANCE / GENOTYPE\_MISS\_RATE): 4  
Recommended: DP=11, GQ=30, LAF=0.30, HAF=0.70



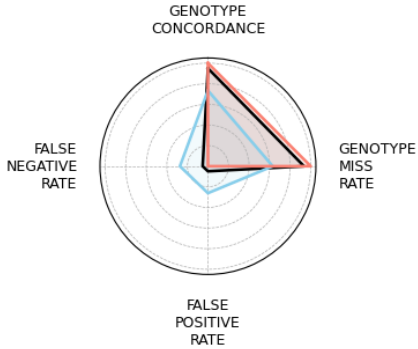
|     | GENOTYPE CONCORDANCE | GENOTYPE MISS RATE | FALSE POSITIVE RATE | FALSE NEGATIVE RATE |
|-----|----------------------|--------------------|---------------------|---------------------|
| ALL | 0.99980              | 0.14480            | 0.00010             | 0.00010             |
| 30X | 0.99978              | 0.10124            | 0.00011             | 0.00011             |
| 15X | 0.99981              | 0.15418            | 0.00010             | 0.00010             |

W(GENOTYPE\_CONCORDANCE / GENOTYPE\_MISS\_RATE): 5  
Recommended: DP=12, GQ=30, LAF=0.30, HAF=0.70



|     | GENOTYPE CONCORDANCE | GENOTYPE MISS RATE | FALSE POSITIVE RATE | FALSE NEGATIVE RATE |
|-----|----------------------|--------------------|---------------------|---------------------|
| ALL | 0.99982              | 0.18642            | 0.00009             | 0.00009             |
| 30X | 0.99979              | 0.13242            | 0.00010             | 0.00010             |
| 15X | 0.99982              | 0.19803            | 0.00009             | 0.00009             |

W(GENOTYPE\_CONCORDANCE / GENOTYPE\_MISS\_RATE): 6  
Recommended: DP=12, GQ=30, LAF=0.30, HAF=0.70



|     | GENOTYPE CONCORDANCE | GENOTYPE MISS RATE | FALSE POSITIVE RATE | FALSE NEGATIVE RATE |
|-----|----------------------|--------------------|---------------------|---------------------|
| ALL | 0.99982              | 0.18642            | 0.00009             | 0.00009             |
| 30X | 0.99979              | 0.13242            | 0.00010             | 0.00010             |
| 15X | 0.99982              | 0.19803            | 0.00009             | 0.00009             |

## Genomic Quality Evaluation Across ALL / 30X / 15X Platforms Under Varying Trade-off Weights

■ ALL ■ 30X ■ 15X

GENOTYPE CONCORDANCE: 9.9971e-01 ~ 9.9982e-01  
GENOTYPE MISS RATE: 2.0622e-02 ~ 1.9803e-01  
FALSE POSITIVE RATE: 8.6163e-05 ~ 1.5773e-04  
FALSE NEGATIVE RATE: 9.2066e-05 ~ 1.3538e-04

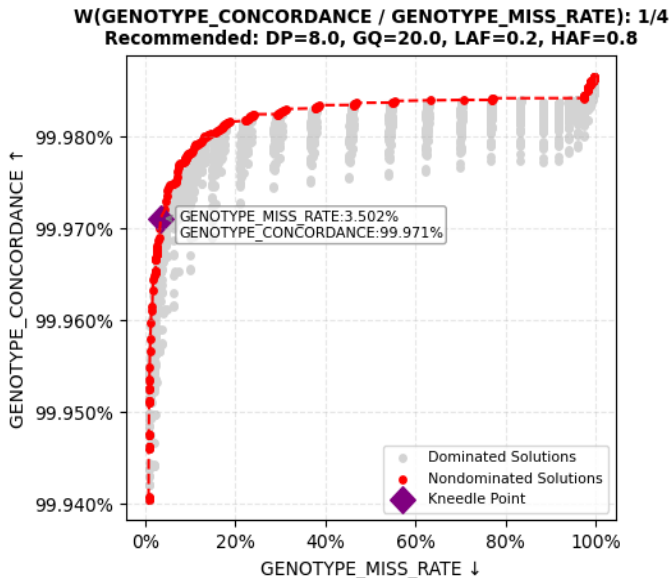
## Result- -Multi-platform Genomic Quality Assessment (4)

Under different reference parameters provided by various weights (W), the main gaps between the ALL, 15X, and 30X platforms are manifested in the following aspects:

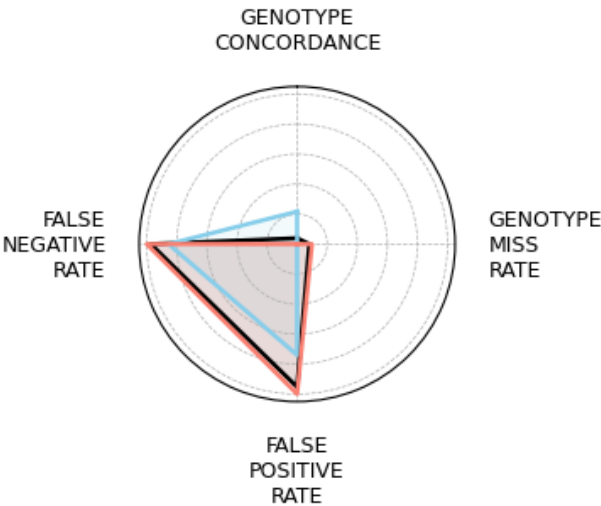
- **Genotype Concordance:**
  - As the W value increases from 1:7 to 6:1, the concordance of all three platforms—ALL, 30X, and 15X—shows slight improvement
  - The 30X platform consistently maintains the highest concordance, increasing from 99.97% to over 99.98%
  - The 15X platform has slightly lower concordance, but remains close to 99.97%
- **Genotype Miss Rate:**
  - As W value increases, the missing rates for all platforms rise significantly
  - The 15X platform consistently has the highest missing rate, reaching 18-19% at high W values
  - The 30X platform has the lowest missing rate, 3-5 percentage points lower than the 15X platform
- **False Positive Rate & False Negative Rate:**
  - As the W value increases from 1:7 to 6:1, both false positive and false negative rates generally show a downward trend
  - The false positive rates between 30X and 15X platforms are quite similar
  - The false negative rates between 30X and 15X platforms are quite similar

# Conclusion

DP: Sequencing Depth  
GQ: Genotype Quality  
AF: Allele Fraction  
LAF: Lower AF  
HAF: Higher AF



W(GENOTYPE\_CONCORDANCE / GENOTYPE\_MISS\_RATE): 1/4  
Recommended: DP=8, GQ=20, LAF=0.20, HAF=0.80



|     | GENOTYPE CONCORDANCE | GENOTYPE MISS RATE | FALSE POSITIVE RATE | FALSE NEGATIVE RATE |
|-----|----------------------|--------------------|---------------------|---------------------|
| ALL | 0.99971              | 0.03502            | 0.00015             | 0.00013             |
| 30X | 0.99973              | 0.02062            | 0.00014             | 0.00013             |
| 15X | 0.99971              | 0.03811            | 0.00016             | 0.00014             |

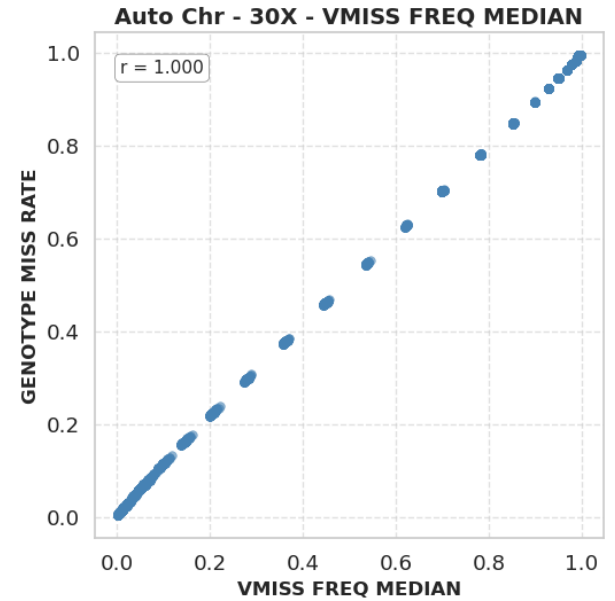
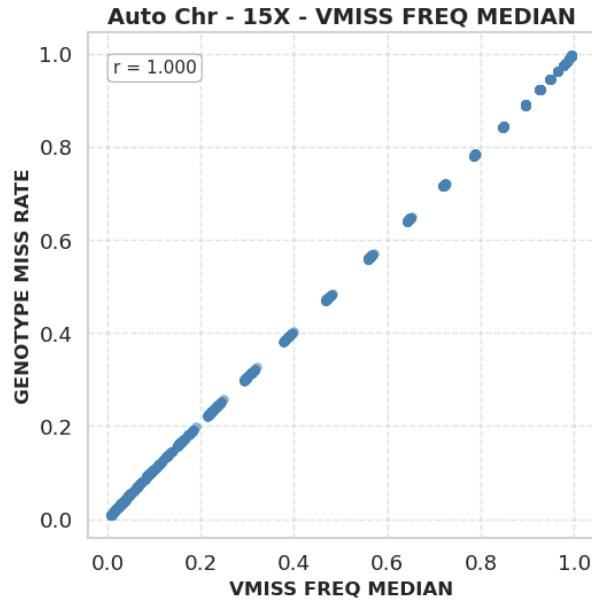
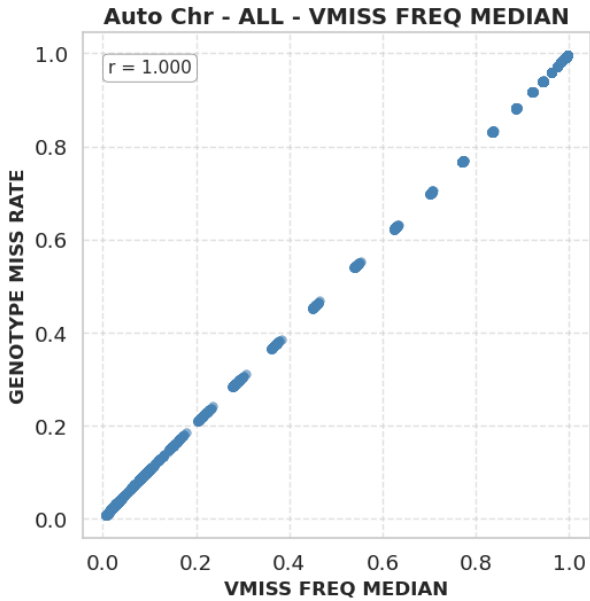
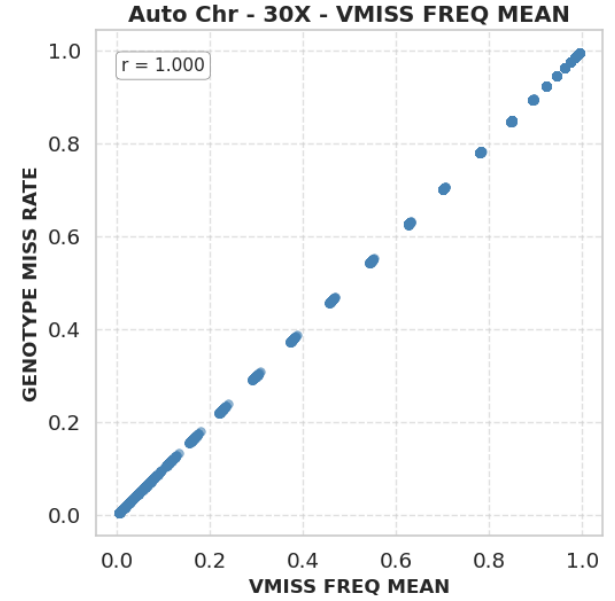
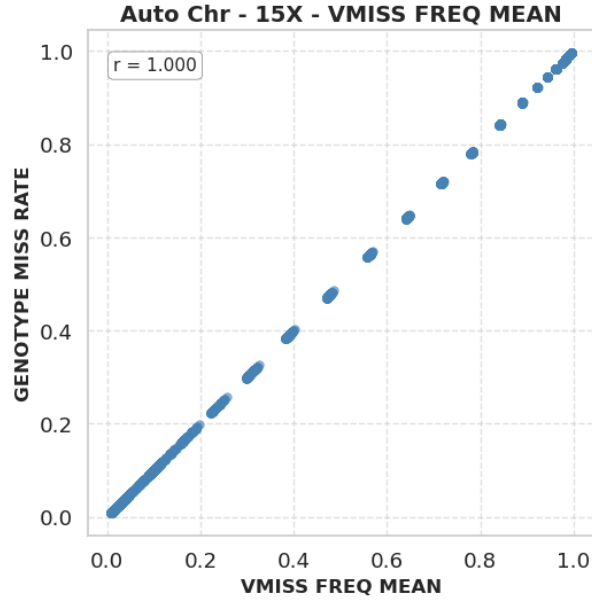
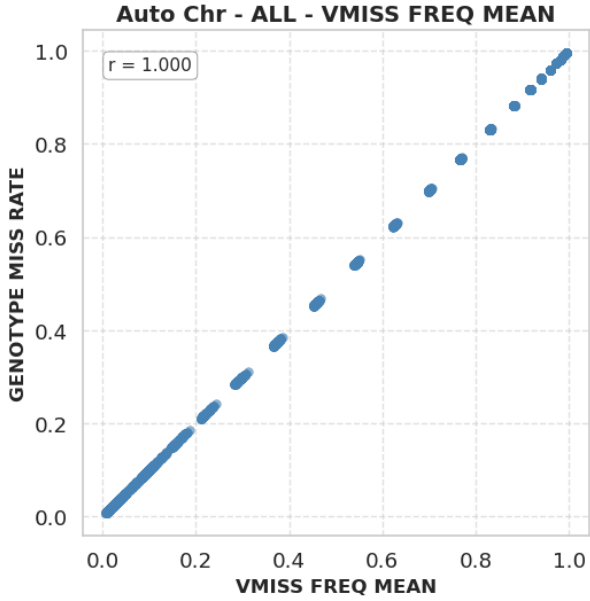
Genomic Quality Evaluation Across ALL / 30X / 15X Platforms  
Under Varying Trade-off Weights  
■ ALL ■ 30X ■ 15X

Based on the comprehensive analysis above, we conclude:  
Recommended parameter combination: **DP = 8, GQ = 20, LAF = 0.20, HAF = 0.80**

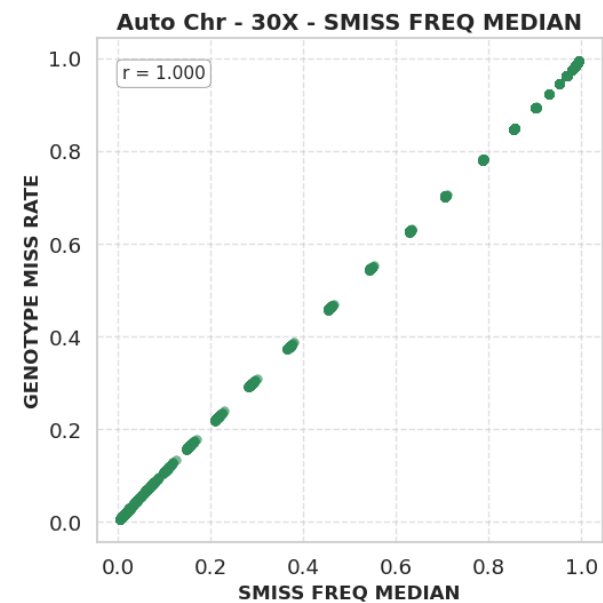
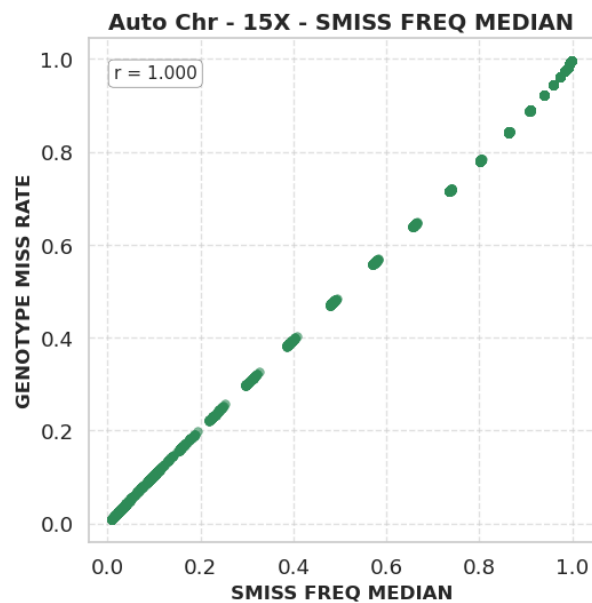
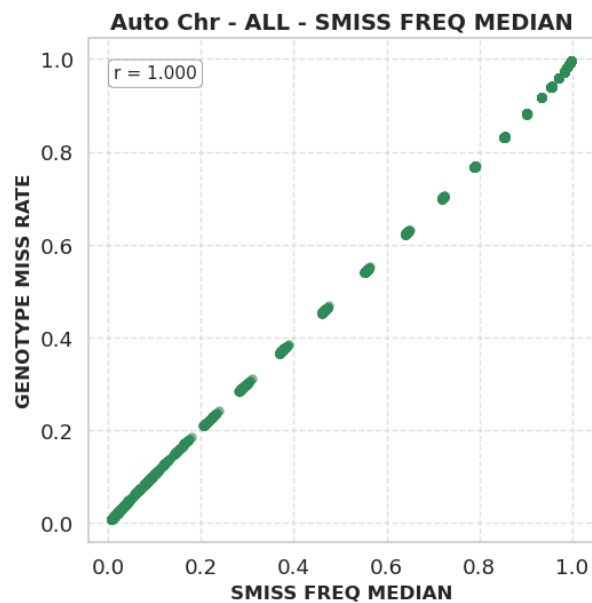
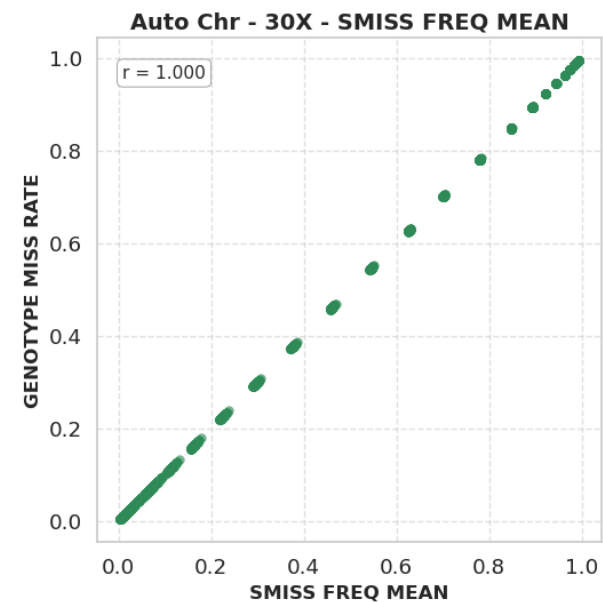
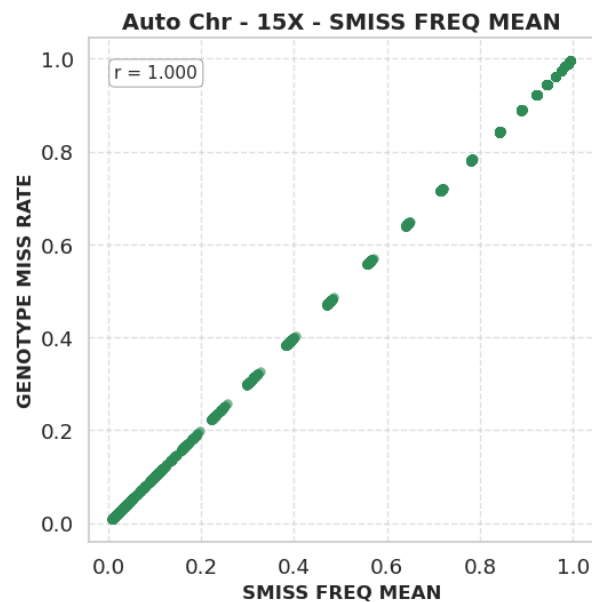
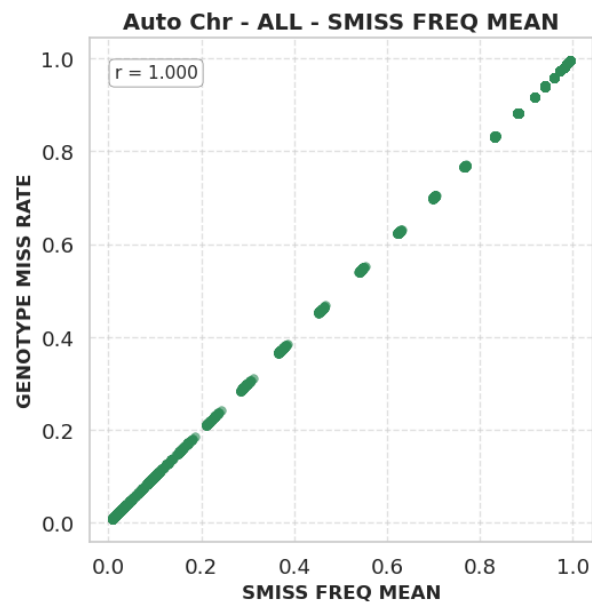
- **Trade-off solution on the Pareto frontier**
  - At  $W=1/7 \sim 1/4$ , the recommended parameters are located on the Pareto Front and identified as the "knee point" by Kneedle detection, indicating this point achieves optimal trade-off between Concordance and Miss Rate
  - **Genotype Concordance** reaches **99.97%**, **Miss Rate** is controlled at **3.50%**, balancing high accuracy with acceptable missing rate.
- **Cross-platform metrics show high consistency (30X / 15X)**
  - Under the recommended parameters, 30X and 15X platforms maintain **high consistency and stability** (Genotype Concordance, False Positive Rate, False Negative Rate)

This parameter combination provides reasonable performance for both 30X and 15X platforms but note that the **15X platform will have a slightly higher missing rate than the 30X platform under these parameters**. Therefore, we need to further examine in subsequent analyses **whether this cross-platform differential missing is unevenly distributed across different genotypes**, as this could lead to potential false positive results.

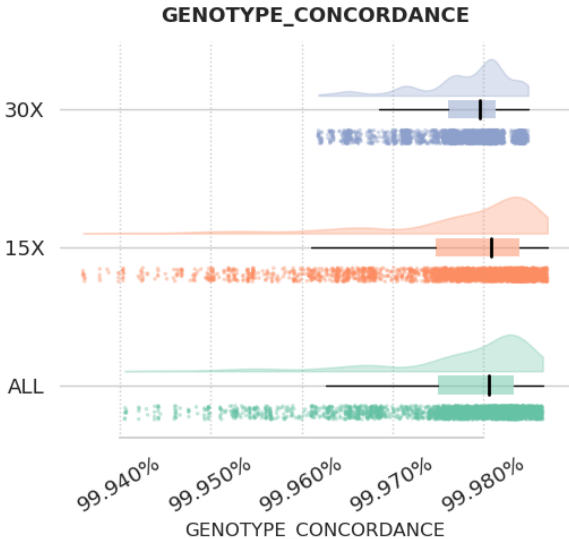
Auto Chr - VMISS vs. GENOTYPE MISSING RATE



## Auto Chr - SMISS vs. GENOTYPE MISSING RATE

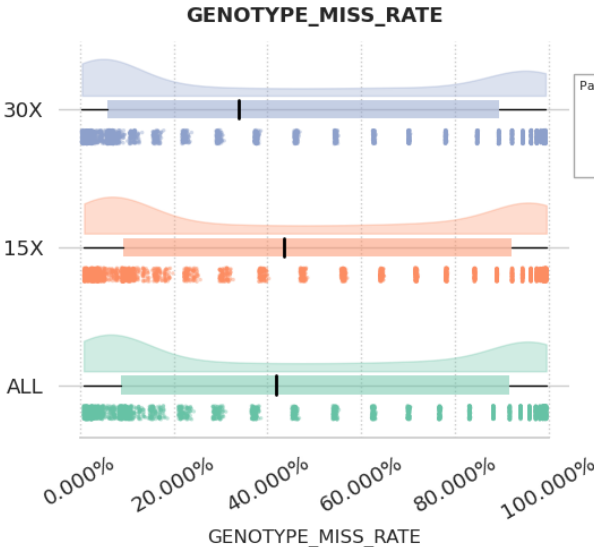


# Cross-Platform Comparison of Genotype Consistency Metrics in Auto Chr



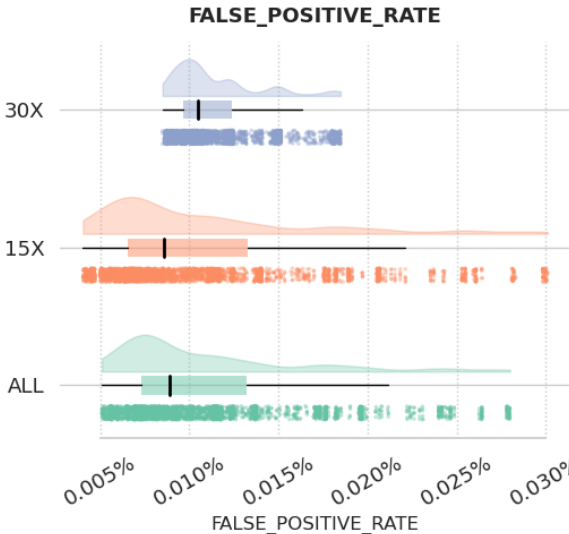
Pairwise comparison (Mann-Whitney U test):

| Group      | $\Delta$ median | $\Delta$ mean | p       |
|------------|-----------------|---------------|---------|
| ALL vs 15X | -2.5e-06        | +9.8e-07      | 1.8e-03 |
| ALL vs 30X | +9.8e-06        | -7.2e-06      | 5.0e-17 |
| 15X vs 30X | +1.2e-05        | -8.2e-06      | 2.0e-25 |



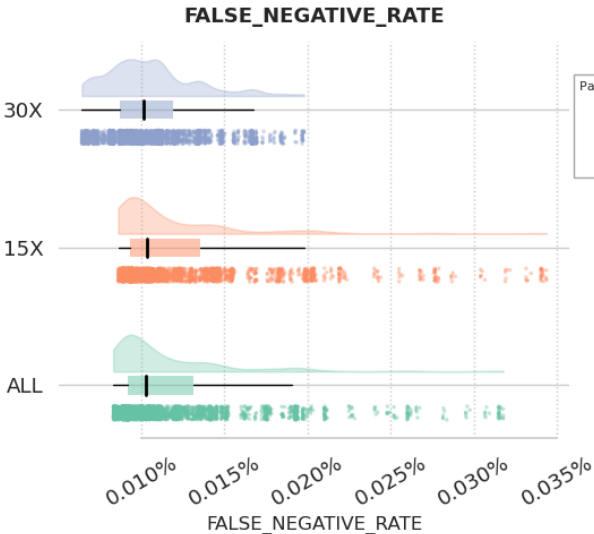
Pairwise comparison (Mann-Whitney U test):

| Group      | $\Delta$ median | $\Delta$ mean | p       |
|------------|-----------------|---------------|---------|
| ALL vs 15X | -1.7e-02        | -7.2e-03      | 2.6e-02 |
| ALL vs 30X | +8.0e-02        | +3.4e-02      | 3.8e-07 |
| 15X vs 30X | +9.7e-02        | +4.1e-02      | 1.3e-10 |



Pairwise comparison (Mann-Whitney U test):

| Group      | $\Delta$ median | $\Delta$ mean | p       |
|------------|-----------------|---------------|---------|
| ALL vs 15X | +3.5e-06        | +2.0e-06      | 6.6e-10 |
| ALL vs 30X | -1.6e-05        | -4.4e-06      | 2.2e-87 |
| 15X vs 30X | -1.9e-05        | -6.3e-06      | 1.6e-92 |



Pairwise comparison (Mann-Whitney U test):

| Group      | $\Delta$ median | $\Delta$ mean | p       |
|------------|-----------------|---------------|---------|
| ALL vs 15X | -7.9e-07        | -2.9e-06      | 2.1e-05 |
| ALL vs 30X | +1.5e-06        | +1.2e-05      | 1.9e-19 |
| 15X vs 30X | +2.2e-06        | +1.5e-05      | 1.4e-31 |